Conflict-Aware Logit Purification for Multi-Teacher Distillation in Low-Resource Text Classification

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Abstract

Combining several large language models as teachers can enrich the supervision available to a compact classifier. However, their outputs may differ in class preference and confidence scale, while querying every teacher for every example is expensive. This dissertation investigates whether these differences can be corrected before aggregation and whether teacher queries can be allocated selectively when both gold labels and teacher calls are limited.

Three frozen instruction-tuned language models supervise a ModernBERT-base student across four text-classification tasks under two few-label regimes. The proposed logit-purification procedure matches each teacher's pool-average prediction to a supplied class prior and then aligns confidence scales. It is evaluated against single-teacher and naive ensemble baselines, alongside a learned per-example router and an entropy-triggered query cascade with cost- and depth-matched random controls.

Pool-level correction improves naive teacher targets across all evaluated tasks and often transfers to better students when the supplied prior is accurate. However, the corrected committee does not establish an advantage over a corrected single teacher, and improvements in target accuracy and calibration do not reliably translate into student performance. Neither learned routing nor selective querying provides a robust benefit in the evaluated low-label regime. Distillation nevertheless remains substantially stronger than label-only supervised training.

These findings show that apparent teacher conflict partly reflects systematic output mismatch rather than irreducible knowledge disagreement. They support a staged approach: correct pool-level mismatch, compare against a corrected single-teacher baseline, and validate each intervention through the trained student before introducing adaptive routing or query allocation.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Ömer Tuğrul*)

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Chapter 1

Introduction

1.1 Motivation

Pre-trained encoder models are widely used for practical text classification, but their effectiveness is often constrained when only a small amount of labelled data is available, which remains a significant practical challenge. Modern instruction-tuned large language models (LLMs) offer a way around this bottleneck: prompted with a task description and a handful of examples, they can produce usable predictions for unlabelled text without any task-specific training [2]. However, they are poorly suited to deployment at scale, since serving a model with tens of billions of parameters for a high-volume classification service is slow and expensive. Knowledge distillation (KD) resolves this tension by using one or more LLMs as "teachers" whose soft predictions supervise the training of a small, deployable "student" [14].

Given access to multiple capable LLMs, it is natural to consider using them together. This is because different models encode different inductive biases, and multi-teacher distillation has shown benefits in image classification [35]. TinyLLM combines rationales from multiple LLM teachers [32]. A later evaluation that expanded its teacher pool found that student accuracy peaked with two teachers and then declined after adding third and fourth teachers across three student sizes [16]. The usual explanation is teacher conflict. When teachers produce contradictory soft predictions for the same input, averaging their logits yields a blurred distribution that conveys neither teacher's knowledge, and the student inherits confusion rather than expertise.

Two practical constraints make this problem particularly relevant and motivate the present work. First, labels are scarce: we assume only $L \in \{10, 50\}$ gold-labelled examples per class, too few to assess teacher quality on a per-sample basis. Second,

teacher queries are not free: each call to a hosted LLM costs money and time, so a budget of $T < KN$ queries over N training samples and K teachers forces a choice about how supervision is allocated. Existing conflict-aware distillation methods largely assume away one or both constraints, requiring labels for every training sample [38], operating on costly natural-language rationales [32], or ignoring query cost altogether. Cost-aware LLM routing addresses budgets, but at inference time rather than during distillation [4, 25]. To the best of our knowledge, no prior work jointly addresses teacher conflict, label scarcity, and query budgets at the logit level for text classification.

1.2 From routing to purification by calibration

Motivated by work on teacher conflict [8] and logit-scale effects in distillation [31], we distinguish two sources of apparent teacher conflict. *Content disagreement* occurs when teachers favour different classes for the same input; *confidence-scale disagreement* occurs when their class rankings are similar but their logit scales differ. This thesis investigates whether distinguishing and addressing these sources of disagreement before aggregation can improve multi-teacher distillation under strict label and query budgets. Our starting hypothesis was that teacher disagreement is structured signal rather than noise: lightweight disagreement features (prediction entropy, pairwise divergence, rank disagreement) should identify cases in which naive averaging produces a harmful teacher target, and a small router trained on the few available labels should exploit them to re-weight teachers per sample. We use *logit purification* as an umbrella term for correcting or selectively combining teacher logits before they supervise the student.

The evidence supports a more limited version of the original approach. In the teacher pool studied here, a substantial fraction of measured conflict is not disagreement about task content, but a confidence-scale artifact: one teacher is substantially more confident than the others, so raw logit averaging gives its predictions disproportionate influence. Two pool-level corrections reduce this artifact without using individual labels during fitting: matching each teacher's average prediction to a supplied class prior, which represents the expected proportion of examples in each class, and aligning teacher confidence scales. The resulting aligned mean outperforms the naive raw-logit mean at the teacher level on every evaluated dataset.

Learned per-sample routing is retained as a negative feasibility result: with the available calibration data, it is rarely selected and does not outperform naive averaging after student training.

Two further findings shape the thesis. First, better teacher-level targets do not automatically produce better students. At $L = 10$, alignment improves both teacher targets and students across all four task configurations, but the sizes of these gains do not correspond: their descriptive Spearman rank correlation is $\rho = 0$. At $L = 50$, reversals occur, most notably on Amazon. Second, an entropy-triggered query cascade that appears favourable in cached-logit simulations shows no consistent advantage over either cost-matched or depth-matched random allocation after student training. Together, these findings show that target-level diagnostics are not sufficient predictors of student-level performance.

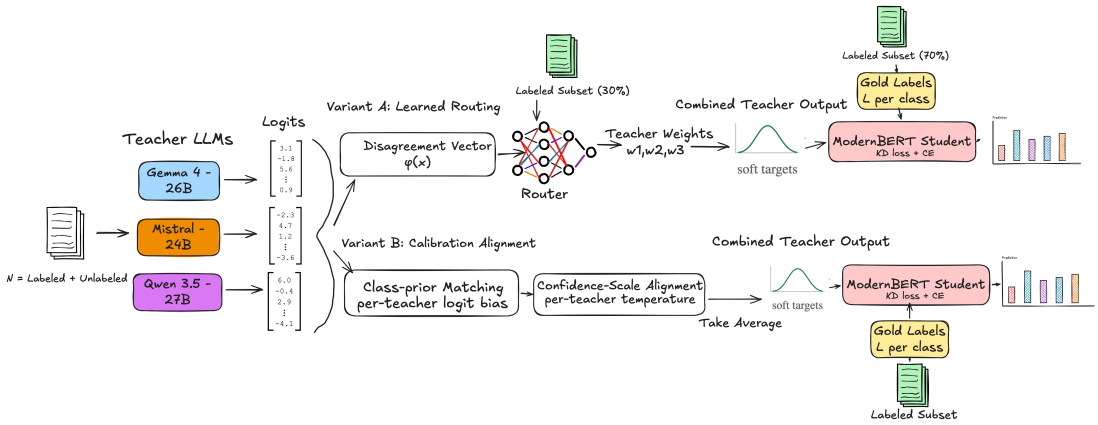


Figure 1.1: Overview of the multi-teacher distillation variants evaluated in this work.

1.3 Research questions and hypotheses

1. **Main RQ.** Can conflict-aware logit purification enable multi-teacher distillation to outperform single-teacher distillation and naive multi-teacher averaging under strict label and teacher-query budgets?

2. **RQ1.** Which disagreement features best identify cases in which at least one teacher predicts the correct class but the aggregate does not, and can a router trained on a small calibration set use these features to improve student performance over naive averaging?

3. **RQ2.** Under a fixed total teacher-query budget T, should the budget be spent on broader coverage with fewer teachers per example, or on deeper supervision with more teachers for a selected subset of examples? How does this choice interact with teacher disagreement?

4. **RQ3.** Do improvements in teacher-target quality transfer to trained-student accuracy and macro-F1, and how does this relationship vary with the label budget?

The hypotheses were specified in the project proposal before the main experiments. H1 predicts that the gain from prior-and-scale alignment over arithmetic probability averaging increases with teacher disagreement. We assess H1 descriptively across tasks at both the teacher-target and trained-student levels; with only four task configurations, the comparison is not treated as an inferential test. H2 predicts that the benefit of distillation over supervised training increases as the label budget decreases; and H3 predicts that the entropy-triggered cascade outperforms equal-cost random allocation. An additional depth-matched control isolates example selection while preserving the cascade's numbers of one-, two-, and three-teacher examples.

1.4 Contributions

This thesis makes five contributions.

1. A logit-level empirical characterisation of three instruction-tuned LLM teachers across four task configurations, distinguishing predictive disagreement from systematic differences in class preference and confidence scale.

2. A two-stage pool-level purification procedure that combines class-prior matching with confidence-scale alignment for black-box teacher logits. The procedure improves naive teacher targets without using individual labels during fitting, although its benefit depends on the supplied class prior (Tables A.11 and 4.5).

3. A trained-student comparison of raw and corrected single-teacher and committee targets. The results show that teacher-target improvements do not reliably transfer to students and do not establish an advantage for the corrected committee over the corrected single-teacher baseline.

4. A feasibility study of learned per-example routing under extreme label scarcity. Full-pool diagnostics reveal potentially useful structure in teacher disagreement, but the gate-controlled student experiment shows that this structure cannot be learned reliably from the available calibration sets.

5. A matched-budget, multi-seed evaluation of entropy-triggered cascading against both cost-matched and depth-matched random allocation. Neither control provides evidence that entropy-based selection improves trained-student performance at the evaluated query budgets.

Code and reproducibility materials. The code, experiment configurations, cached analysis outputs, and reproduction instructions used in this dissertation are available at `https://github.com/omer38/Conflict-Aware-Logit-Purification`.

1.5 Thesis outline

Chapter 2 reviews the background: multi-teacher distillation and teacher conflict, budget-aware and low-label distillation, and the calibration literature this work draws on. Chapter 3 defines the problem setting, datasets, teacher pool and student architecture, and describes the purification, routing and cascade methods together with the evaluation protocol. Chapter 4 presents the experiments in a logical sequence. It first diagnoses teacher conflict, class bias, and confidence-scale differences; then evaluates aligned aggregation at the teacher level; next tests whether those target-level gains transfer through distillation to students; and finally evaluates routing and matched-budget query allocation. Chapter 5 discusses what the results imply for multi-teacher distillation practice, revisits the hypotheses, and states the limitations of the study. Chapter 6 concludes and outlines directions for future work, including evaluation with genuinely heterogeneous teacher pools in which routing may prove more useful.

Chapter 2

Background and Related Work

2.1 Knowledge Distillation and Disagreement Measures

Knowledge distillation trains the student to match a teacher-derived target $P_T(x)$, with cross-entropy supervision added for labelled examples [14]. Let C denote the number of classes. Teacher k produces logits $z_k(x) \in \mathbb{R}^C$, while the student f_θ produces logits $z_\theta(x)$. Temperature-scaled probabilities are

$$\sigma_\tau(z)_c = \frac{\exp(z_c/\tau)}{\sum_{m=1}^{C} \exp(z_m/\tau)}.$$

Increasing τ softens the distribution without changing its class ordering. Unlike a one-hot label, $P_T(x)$ retains information about non-maximum classes. Section 3.9 specifies the exact training objective used in the experiments.

For teacher probabilities $p_k(x) = \sigma_1(z_k(x))$, define the probability-space consensus as

$$\bar{p}(x) = \frac{1}{K} \sum_{k=1}^{K} p_k(x).$$

We measure individual and ensemble uncertainty using Shannon entropy, $H(p_k(x))$ and $H(\bar{p}(x))$, respectively. Pairwise disagreement is measured by the symmetrised KL divergence, $D_{ij}(x) = \frac{1}{2}\left[D_{\mathrm{KL}}(p_i(x)\|p_j(x)) + D_{\mathrm{KL}}(p_j(x)\|p_i(x))\right]$, while $D_{\mathrm{KL}}(p_k(x)\|\bar{p}(x))$ measures each teacher's deviation from the consensus. Because probability divergence can be large even when class rankings agree, we additionally use top-1 agreement and Kendall's τ rank correlation [17].

2.2 Knowledge Distillation under Label Scarcity

Instruction-tuned LLMs can provide soft supervision for unlabelled text through prompting [2], making distillation useful when gold labels are scarce. Existing low-label methods primarily assume one teacher. COD generates counterfactual examples near the teacher's decision boundary [12], PromptMix generates and relabels borderline examples [29], and performance-guided distillation adapts teacher data generation using student validation performance [26]. These methods establish the value of LLM supervision in low-label settings, but do not determine how several teachers should be combined when they disagree.

Teacher quality and student teachability must also be distinguished. More accurate or higher-capacity teachers do not necessarily produce better students when the teacher–student capacity gap makes their predictions difficult to imitate [6]. Moreover, teacher–student fidelity and student generalisation are separate outcomes: closer matching of the teacher distribution does not always improve test performance [30].

2.3 Multi-Teacher Distillation and Teacher Conflict

A standard multi-teacher baseline averages softened teacher predictions [35]. Adaptive methods instead vary teachers by example. CA-MKD weights teachers through their losses against the gold label [38]. RL-KD uses reinforcement learning to select teachers per instance from input representations, teacher probabilities and label-dependent teacher losses, updating the policy through subsequent student performance [36]. It is a direct NLP precedent for instance-specific routing, but requires per-example labels and iterative joint student–selector training rather than a separable router fitted from a small calibration set.

Disagreement alone does not imply that combining teachers will improve performance. Ensemble theory characterises diversity as part of a bias–variance–diversity trade-off rather than an objective that should be maximised independently; for classification accuracy, its effect can also depend on the label distribution [34]. Useful complementarity therefore requires disagreement on cases where another teacher supplies recoverable information, not merely a large divergence between predictions.

Label-free alternatives use confidence or student optimisation. Entropy-based methods favour confident teachers [5, 18], which may fail when low entropy reflects overconfidence. AE-KD seeks an update direction compatible with multiple teacher

losses [8], while GOVERN selects teachers whose student-gradient orientations agree with the majority [44]. These methods motivate adaptive selection but do not correct pool-level class-marginal or confidence-scale mismatch before aggregation.

Recent LLM-to-LLM work studies conflict in generated answers and rationales, motivating purification and routing [32, 16]. PerSyn instead routes each prompt according to teacher response quality and student learnability [39]. These methods operate on generated responses, whereas the present work studies constrained class logits under a fixed query budget.

Logit-refinement research nevertheless shows that output geometry matters. DKD separates target- and non-target-class information [41], while logit standardisation decouples teacher and student scales [31]. PILE instead learns a supervised multi-teacher logit ensemble [3]. The present work addresses a different interface: scale mismatch between teachers before their outputs supervise the student.

2.4 Budget-Aware Distillation and Cost-Aware Routing

Budget-aware distillation has mainly considered whether to query a single teacher for a given example. M-RARU uses uncertainty-triggered querying [20], while performance-guided and robust active distillation also allocate supervision across examples [26, 1]. They do not address how a fixed budget should be divided across several teachers.

At inference time, FrugalGPT uses a cheap-to-expensive cascade [4], and RouteLLM learns a lightweight model router [25]. Their success does not directly imply that the same allocation will produce the best distilled student: additional queries affect a training target rather than the immediate prediction. This motivates the trained-student comparison of multi-teacher allocation policies in Chapter 4.

2.5 Calibration of Prompted Classifiers

Predictive confidence can differ substantially between neural models, and temperature scaling can correct confidence without changing the predicted class [11]. Calibration also affects knowledge transfer [9]. When several teachers are averaged in logit space, differences in logit scale additionally change their relative influence.

The aggregation operator determines how these differences enter the consensus. Arithmetic probability averaging is a linear opinion pool, whereas averaging logits and applying softmax produces a logarithmic opinion pool, equivalent to a normalised

geometric mean of teacher probabilities [13]. The latter can strongly suppress a class when one teacher assigns it very low probability, while confidence-scale differences create unequal effective influence despite nominally uniform weights. This connects the choice of aggregation space directly to the need for confidence-scale alignment.

Prompted classifiers can exhibit biases from prompt format, verbalizer choice, example order, or batch composition, motivating contextual and batch calibration [42, 43]; related class-prior and logit adjustments address prior shift and class imbalance [28, 22]. Following this principle, the method in Chapter 3 fits teacher-specific offsets so that each teacher's pool-average prediction matches a supplied prior, then aligns their confidence scales. The procedure uses no individual labels during fitting, but is unsupervised only conditional on the supplied prior.

2.6 Aggregating Noisy Annotators

The Dawid–Skene model treats true labels as latent and uses expectation–maximisation to estimate each annotator's class-conditional confusion matrix [7]. In this thesis, teachers are treated as annotators, and their probability vectors replace hard categorical votes. The resulting soft Dawid–Skene-inspired posterior is used only as a diagnostic of class-conditional expertise, not as a principal aggregation method.

2.7 Research Gap and Positioning

Existing work separately studies low-label single-teacher distillation, multi-teacher weighting, calibration, and cost-aware routing. Their intersection remains underexplored: multiclass text distillation with several black-box LLM teachers, very few gold labels, and a fixed query budget.

This thesis addresses that intersection through three linked components: pool-level correction of class-marginal and confidence-scale mismatch, few-example routing from disagreement features, and selective allocation of additional teacher queries. It evaluates each component through trained students rather than relying only on teacher-target metrics. A detailed comparison between the implemented baselines and existing methods appears in Table A.13.

Chapter 3

Methods and Experimental Design

3.1 Problem Formulation

We study multiclass text classification under limited gold-label and teacher-query budgets. Let

$$\mathcal{D} = \{x_i\}_{i=1}^{N}$$

be a working pool of texts with label space $\mathcal{Y} = \{1, \ldots, C\}$, and let $I_L \subseteq \{1, \ldots, N\}$ index the examples whose labels are revealed. The class-balanced labelled and unlabelled sets are

$$\mathcal{D}_L = \{(x_i, y_i) : i \in I_L\}, \qquad |\{(x_i, y_i) \in \mathcal{D}_L : y_i = c\}| = L \quad \forall c \in \mathcal{Y},$$

and

$$\mathcal{D}_U = \{x_i : i \notin I_L\}.$$

Consequently, $|\mathcal{D}_L| = LC$ and $|\mathcal{D}_U| = N - LC$. We evaluate the low-resource regimes $L \in \{10, 50\}$ labels per class.

A pool of K frozen teacher LLMs is available. For a prompted input x, teacher k returns

$$z_k(x) = [\log p_k(v_1 \mid x), \ldots, \log p_k(v_C \mid x)] \in \mathbb{R}^C,$$

where v_c is the verbalizer for class c. These class-token log-probabilities can be used as logits because the softmax is invariant to a common additive constant. Guided decoding ensures that all C verbalizer probabilities are returned.

Teacher access is represented by a query set

$$Q \subseteq \{1, \ldots, N\} \times \{1, \ldots, K\},$$

where $(i,k) \in Q$ means that teacher k is queried for x_i. Define

$$B(Q) = |Q|, \qquad q(Q) = \frac{B(Q)}{N}, \qquad \mathcal{K}_i = \{k : (i,k) \in Q\}.$$

Here, $B(Q)$ is the total query cost, $q(Q)$ is the mean number of queries per example, and $\mathcal{K}_i$ contains the teachers queried for x_i. Policies query at least one teacher per example and must satisfy

$$N \le B(Q) \le KN, \qquad 1 \le q(Q) \le K, \qquad B(Q) \le T,$$

where T is the teacher-query budget.

A target-construction policy combines the query allocation Q with an aggregation rule $\mathcal{A}$. Using only the queried outputs and permitted side information, it constructs

$$P_{\mathcal{A},Q}(x_i) = \mathcal{A}\big(\{z_k(x_i) : k \in \mathcal{K}_i\}, \phi(x_i); \pi\big),$$

where $P_{\mathcal{A},Q}(x_i)$ is a probability vector, $\phi(x_i)$ denotes optional disagreement features, and π is an optional supplied class prior. The rule may implement fixed aggregation, input-dependent routing, or teacher-specific calibration. Because the working pools are class-balanced, prior-matching experiments use $\pi = (1/C, \ldots, 1/C)$. Fitting teacher-specific offsets uses this pool-level prior but no individual labels; the procedure is therefore label-free conditional on the supplied prior.

The encoder-based student f_θ, with logits $z_\theta(x)$, is trained to match these targets using the objective in Section 3.9, Equation 3.20. Distillation is applied to every example in $\mathcal{D}_L \cup \mathcal{D}_U$. Gold-label cross-entropy is applied to all of $\mathcal{D}_L$ in non-router experiments. In router experiments, a stratified subset $\mathcal{D}_{\text{cal}} \subset \mathcal{D}_L$ is reserved for router fitting and model selection, and cross-entropy is applied to

$$\mathcal{D}_{L\text{-tr}} = \mathcal{D}_L \setminus \mathcal{D}_{\text{cal}}.$$

The objective is to select $(\mathcal{A}, Q)$ to maximise the generalisation of the trained student on a disjoint test set, $\mathcal{D}_{\text{test}} \cap \mathcal{D} = \varnothing$, subject to the label and query budgets:

$$\max_{\mathcal{A},\,Q} \quad \text{Perf}\Big(f_{\hat{\theta}_{\mathcal{A},Q}}; \mathcal{D}_{\text{test}}\Big) \qquad \text{subject to} \qquad |\mathcal{D}_L| = LC, \quad B(Q) \le T,$$

where $\hat{\theta}_{\mathcal{A},Q}$ denotes the resulting student parameters. Performance is measured using accuracy and macro-F1. This formulation distinguishes aggregation, which determines how queried outputs form a soft target, from allocation, which determines the teacher–example pairs queried under T.

Table 3.1: Class-balanced distillation pools and fixed test sets; N is the pool size and Amazon-3 is derived from Amazon Reviews.

Dataset	Task	C	N	Test size	Verbalizers
AG News	news topic	4	6,000	7,600	*A*–*D*
Yahoo! Answers	question topic	10	5,000	10,000	*A*–*J*
Amazon Reviews Full	ordinal star rating	5	2,500	10,000	*A*–*E*
Amazon-3	grouped review sentiment	3	1,500	6,000	derived from *A*–*E*

All $K \times N$ teacher outputs are cached, and lower-budget policies are reconstructed by masking teacher–example pairs. Reported costs are therefore counterfactual online query counts; the experiments measure allocation effects on student learning rather than realised latency or monetary savings.

3.2 Datasets and Experimental Splits

3.2.1 Datasets

Table 3.1 summarises the evaluated task configurations. Three are English multiclass text-classification benchmarks from Zhang et al. [40]: AG News, Yahoo! Answers Topics, and Amazon Reviews Full. These respectively represent four-class news-topic, ten-class question-topic, and five-class ordinal-rating tasks. For Amazon Reviews, the title and body are concatenated as input.

We also derive *Amazon-3* as a secondary robustness task by grouping the Amazon Reviews Full labels into negative (one–two stars), neutral (three stars), and positive (four–five stars). It is a coarser representation of the same review data, not an independent benchmark or a measurement of the teachers' native three-class behaviour.

Teacher prompts use single-token verbalizers: *A*–*D* for AG News, *A*–*J* for Yahoo! Answers, and *A*–*E* for Amazon Reviews. These were verified as single tokens for all teacher tokenizers, enabling complete C-class probability extraction through guided decoding.

All working pools and test sets are class-balanced, preventing class prevalence from confounding aggregation comparisons and making the uniform prior used for class-prior matching known by construction.

Amazon-3 reuses the selected Amazon Reviews Full examples and cached teacher

outputs. A teacher's five-class distribution is grouped as

$$p_k^{(3)}(x) = \left[p_{k,1}^{(5)}(x) + p_{k,2}^{(5)}(x),\ p_{k,3}^{(5)}(x),\ p_{k,4}^{(5)}(x) + p_{k,5}^{(5)}(x)\right],$$

and the corresponding score vector is the logarithm of this distribution. Amazon-3 therefore requires no additional teacher queries and isolates the effect of label grouping without changing the texts or teacher pool.

3.2.2 Working-pool construction and low-label splits

For each primary dataset, a fixed class-balanced working pool is sampled from the original training split and a separate class-balanced test set from the original test split. Teacher outputs are cached for the working pool, while the test set remains fixed across methods and seeds.

For each label budget L and training seed, $\mathcal{D}_L$ is sampled without replacement and stratified by class. It always includes two fixed prompt-demonstration examples per class, with the remaining $L-2$ selected randomly. Thus, every label used in a teacher prompt is counted within the declared budget. Each prompt uses one demonstration per class; if the scored example is itself a demonstration, the other fixed example from that class is used to prevent self-demonstration.

All examples in $\mathcal{D}_L$ contribute to cross-entropy training in supervised and non-router experiments. Router experiments instead reserve a stratified calibration subset:

$$|\mathcal{D}_{\text{cal}} \cap \{y = c\}| = \text{round}(0.3L), \qquad \mathcal{D}_{L\text{-tr}} = \mathcal{D}_L \setminus \mathcal{D}_{\text{cal}}.$$

At $L = 10$, this leaves seven cross-entropy and three router-calibration examples per class. The router uses $\mathcal{D}_{\text{cal}}$ for fitting, leave-one-out model selection, and its safety gate. Features are standardised over the full working pool without using labels. These examples are excluded from the student's cross-entropy loss but may still contribute to distillation through their cached teacher targets. Examples in $\mathcal{D}_U$ contribute through distillation only.

Primary student comparisons use seeds $\{42, 123, 456\}$. The matched-budget allocation comparison at $q = 1.4$ additionally uses seeds 789 and 2024, giving five paired replicates, whereas the comparison at $q = 2.0$ uses the three primary seeds. Each seed controls both the stratified labelled draw and stochastic student training; the test set remains fixed. Test labels are never used for aggregation fitting, router selection, temperature fitting, or early stopping.

3.3 Teacher and Student Models

The teacher pool comprises three frozen instruction-tuned LLMs: Qwen3.5-27B [27], Gemma-4-26B-A4B-IT [10], and Mistral-Small-3.2-24B-Instruct-2506 [23]. Teacher logits were extracted using vLLM 0.23.0 [19] on a single NVIDIA RTX PRO 6000 GPU with 96 GB VRAM. The teachers are treated as black-box classifiers: only constrained-decoding class-token log-probabilities are used, without access to parameters, hidden states, or gradients.

The student is ModernBERT-base [33] with a task-specific classification head. Its fixed training configuration is reported in Section 3.9 and Appendix D.

3.4 Teacher Logit Extraction

Each teacher was queried once for every working-pool example, and its outputs were cached for all subsequent experiments. Teachers received the same prompt structure: the task categories, a single-token output instruction, and one demonstration per class. Demonstrations followed the prompt-inclusive and self-demonstration controls defined in Section 3.2; the full template is given in Appendix C.

Inference was performed sequentially through vLLM at temperature 0. Guided decoding restricted the first generated token to the verified single-token verbalizers, producing the complete class-score vector $z_k(x_i)$ defined in Section 3.1. This avoids missing-class imputation and gives every aggregation method the same label space.

The cached outputs were reused for aggregation, calibration, and student training. Lower-budget policies were reconstructed by masking teacher–example pairs, as described in Section 3.1. Algorithm 1 provides the complete extraction procedure.

3.5 Aggregation Baselines

We compare four label-free target-construction baselines that differ in their aggregation space and teacher weighting. For any non-empty queried set $\mathcal{K}_i$, define

$$p_k^{(\tau)}(x_i) = \mathrm{softmax}(z_k(x_i)/\tau), \qquad p_k(x_i) = p_k^{(1)}(x_i),$$

where the KD temperature τ is fixed across methods.

B1: single teacher. A student trained from teacher k receives

$$P_k(x_i) = \text{softmax}\left(\frac{z_k(x_i)}{\tau}\right). \tag{3.1}$$

We train one student per teacher and report the best result as a single-teacher reference. Because this choice is not available during target construction, it is not a deployable selection policy.

B2: uniform logit mean. For a non-empty queried teacher set $\mathcal{K}_i$, uniform logit averaging constructs the target

$$P_{\text{logit}}(x_i) = \text{softmax}\left(\frac{1}{|\mathcal{K}_i|\tau}\sum_{k\in\mathcal{K}_i} z_k(x_i)\right). \tag{3.2}$$

Equivalently, its probability for class c satisfies

$$P_{\text{logit}}(x_i)_c \propto \left(\prod_{k\in\mathcal{K}_i} p_{k,c}(x_i)\right)^{1/(|\mathcal{K}_i|\tau)}. \tag{3.3}$$

Logit averaging therefore forms a softened, normalised geometric mean. It is conjunctive: a teacher assigning very little probability to a class can suppress that class despite support from the other teachers. Although its nominal weights are uniform, teachers with greater classwise logit spread exert more influence. The rule is consequently sensitive to confidence-scale differences but invariant to a common additive shift of a teacher's logits.

B3: arithmetic probability mean. This baseline first converts each teacher's raw logits to a temperature-one probability distribution and averages those distributions arithmetically. The KD temperature is then applied once to the aggregate:

$$\bar{p}(x_i) = \frac{1}{|\mathcal{K}_i|}\sum_{k\in\mathcal{K}_i} \text{softmax}(z_k(x_i)), \qquad P_{\text{prob}}(x_i) = \text{softmax}\left(\frac{\log\max(\bar{p}(x_i), 10^{-12})}{\tau}\right). \tag{3.4}$$

The elementwise clamp makes the logarithm numerically safe. When $\tau = 1$, $P_{\text{prob}}(x_i)$ recovers $\bar{p}(x_i)$ up to any numerical effect introduced by the clamp.

Comparing B2 and B3 therefore isolates the effect of aggregation space before the calibration corrections of Section 3.6. We implement only the output-level component of You et al. [35], because their intermediate-representation constraints require access to teacher hidden states, which is unavailable in the black-box setting.

B4: exponential entropy-weighted logit mean. We additionally evaluate a label-free entropy-weighted baseline inspired by Kwon et al. [18] and Cheng et al. [5]. It uses the output-level information available from the black-box teachers and is not a reproduction of either method.

$$w_k^{\text{ent}}(x_i) = \frac{\exp\left[-H\left(p_k^{(\tau)}(x_i)\right)\right]}{\sum_{j\in\mathcal{K}_i}\exp\left[-H\left(p_j^{(\tau)}(x_i)\right)\right]}, \tag{3.5}$$

$$P_{\text{ent}}(x_i) = \text{softmax}\left(\frac{1}{\tau}\sum_{k\in\mathcal{K}_i} w_k^{\text{ent}}(x_i) z_k(x_i)\right). \tag{3.6}$$

Equal teacher entropies recover B2. Otherwise, a sharper teacher is favoured both through its entropy weight and through its greater logit spread. Consequently, B4 assumes that lower entropy indicates greater reliability, which may fail when teachers are differently calibrated. Entropies are computed at the KD temperature τ; the router in Section 3.7 instead constructs its entropy prior at $\tau = 1$.

None of these baselines corrects class-marginal bias or confidence-scale differences. Section 3.6 introduces those corrections.

3.6 Prior and Confidence-Scale Alignment

Raw-logit aggregation is sensitive to two pool-level distortions: class-marginal bias and differences in teacher confidence scale. Neither necessarily represents content disagreement. We therefore apply three ordered stages: matching each teacher's marginal prediction to a supplied class prior, aligning mean predictive entropy across teachers, and averaging the corrected distributions arithmetically. Parameters are fitted on the working pool without using individual labels or test data. Let $\pi \in \Delta^{C-1}$, with $\pi_c > 0$, denote the supplied prior. Algorithm 2 gives the complete procedure.

Stage 1: class-prior matching. The first stage aims to remove systematic class preference by shifting each teacher's logits so that its average predicted distribution matches the supplied class prior. For teacher k, the pool marginal induced by an additive offset $b_k \in \mathbb{R}^C$ is

$$\hat{\pi}_k(b_k) = \frac{1}{N}\sum_{x\in\mathcal{D}} \text{softmax}\big(z_k(x) + b_k\big). \tag{3.7}$$

We seek $\hat{\pi}_k(b_k) = \pi$ using the fixed-point update

$$b_k \leftarrow b_k + \log\pi - \log\hat{\pi}_k(b_k), \tag{3.8}$$

terminated when $\|\Delta b_k\|_\infty < \varepsilon$ or after M iterations.

Softmax shift invariance makes b_k identifiable only up to a common constant, without affecting corrected probabilities. The class-specific offset may change top-1 predictions. Although global convergence is not claimed, every fixed point satisfies the required moment condition. The procedure is related to prior adjustment [28] and contextual calibration [42].

Stage 2: confidence-scale alignment. The second stage reduces confidence-scale differences by flattening sharper teachers until their pool-level mean entropy matches that of the least-confident teacher. After prior matching, define teacher k's mean entropy at scale t as

$$\bar{H}_k(t) = \frac{1}{N} \sum_{x \in \mathcal{D}} H\left(\text{softmax}\left(\frac{z_k(x) + b_k}{t} \right) \right), \tag{3.9}$$

and set $H_k = \bar{H}_k(1)$. Using the least-confident teacher as the reference, $H^\star = \max_k H_k$, we choose t_k by bisection such that

$$\bar{H}_k(t_k) = H^\star. \tag{3.10}$$

For $\beta = 1/t$ and $p_\beta \propto \exp(\beta z)$,

$$\frac{\mathrm{d}H}{\mathrm{d}\beta} = -\beta \operatorname{Var}_{p_\beta}(z) \leq 0. \tag{3.11}$$

Consequently, $\bar{H}_k(t)$ is continuous and strictly increasing in t when at least one pool example has non-constant logits, giving a unique solution under this non-degeneracy condition. Since $H_k \leq H^\star$, each $t_k \geq 1$: the procedure only flattens sharper teachers.

Positive scalar scaling preserves top-1 predictions, so this stage cannot change individual-teacher accuracy; it changes only relative influence during aggregation. Aligning the pool-level mean also preserves within-teacher variation in per-example uncertainty. Unlike supervised temperature scaling [11], no labels or accuracy-based objective are used; the procedure instead reconciles teacher scales with one another, similarly in spirit to logit standardisation [31].

Stage 3: aggregation. The final stage applies baseline B3 after the two teacher-specific corrections: it averages the corrected distributions in probability space and then applies the KD temperature once to the aggregate. This avoids the geometric-mean veto of Equation 3.3. The corrected distribution of teacher k is

$$\tilde{p}_k(x_i) = \text{softmax}\left(\frac{z_k(x_i) + b_k}{t_k} \right). \tag{3.12}$$

For the queried set $\mathcal{K}_i$, the final target is

$$\bar{p}^{\mathrm{al}}(x_i) = \frac{1}{|\mathcal{K}_i|} \sum_{k \in \mathcal{K}_i} \tilde{p}_k(x_i), \qquad P_{\mathrm{align}}(x_i) = \mathrm{softmax}\left(\frac{\log \max\left(\bar{p}^{\mathrm{al}}(x_i), 10^{-12}\right)}{\tau} \right). \tag{3.13}$$

The stages do not commute, so all combined results apply confidence-scale alignment after prior matching.

Variants. Equation 3.13 reduces to B3 when $b_k = \mathbf{0}$ and $t_k = 1$, to prior-matched averaging when $t_k = 1$, and to scale-aligned averaging when $b_k = \mathbf{0}$. Prior matching may change an individual teacher's prediction; scale alignment preserves each teacher's argmax but can change the aggregate by altering relative influence. We also evaluate a prior-matched single teacher to separate correction from aggregation.

Assumptions and cost. Stage 1 is label-free conditional on the supplied uniform prior, which is known because the working pools are balanced; sensitivity to this assumption is reported in Table A.11. Parameters are fitted and applied transductively without individual labels. Under partial querying, they can be estimated only from outputs available for each teacher, as examined in Table A.12. Both stages cost $O(KNC)$ per iteration and require no additional teacher queries or gradients.

Student-level prior sensitivity. To test whether prior sensitivity survives distillation, we retain the balanced working and test pools and vary only the prior supplied to alignment. Mild and strong misspecification use $\pi^{(s,0)} = (1-s)\pi_{\mathrm{uniform}} + se_0$, with $s \in \{0.15, 0.50\}$ and class 0 predeclared as dominant. We evaluate AG News, Yahoo and Amazon at $L \in \{10, 50\}$ using seeds $\{42, 123, 456\}$, giving 54 cells including the existing correct-prior baselines. Unlike Table A.11, this student experiment fixes one misspecification direction rather than averaging over all classes.

3.7 Conflict-Aware Router

The router learns example-specific teacher weights from disagreement patterns. It is the only target-construction method fitted with gold labels, using a stratified calibration subset $\mathcal{D}_{\mathrm{cal}} \subset \mathcal{D}_L$. Its eleven input features comprise consensus entropy, pairwise symmetric KL statistics, top-1 agreement, rank agreement, variance on the three leading consensus classes, and each teacher's divergence from the consensus. The complete

definitions are reported in Table B.1 in the appendix. Features are standardised over the working pool without using labels.

The router learns a residual correction to a fixed entropy prior computed at $\tau = 1$. This prior is related to B4, but B4 computes its entropy weights at the KD temperature. For $p_k^{(1)}(x) = \mathrm{softmax}(z_k(x))$,

$$w_k^{\mathrm{prior}}(x) = \frac{\exp[-H(p_k(x))]}{\sum_{j=1}^{K} \exp[-H(p_j(x))]}, \qquad w(x) = \mathrm{softmax}\Big(r_{\theta_r}(\tilde{\phi}(x)) + \log w^{\mathrm{prior}}(x)\Big). \tag{3.14}$$

The resulting distillation target is

$$P_{\mathrm{router}}(x) = \mathrm{softmax}\left(\frac{1}{\tau}\sum_{k=1}^{K} w_k(x) z_k(x)\right). \tag{3.15}$$

Setting the learned residual to zero recovers the fixed $\tau = 1$ entropy-prior logit mean.

We evaluate a linear router and a two-layer MLP with hidden width 16. Each architecture and weight-decay value is assessed by leave-one-out cross-validation on $\mathcal{D}_{\mathrm{cal}}$. The router is deployed only when its best cross-validated accuracy strictly exceeds both the uniform-logit and entropy-weighted baselines; otherwise, the stronger fixed baseline is used. Both architectures are evaluated on AG News, Yahoo, and Amazon at $L = 10$ for seeds $\{42, 123, 456\}$, giving 18 gate-controlled student runs. Algorithm 3 gives the complete fitting and safety-gate procedure.

At $L = 10$, the calibration sets contain only 12, 30, and 15 examples on AG News, Yahoo, and Amazon, respectively. Leave-one-out accuracy therefore changes in increments of 8.3, 3.3, and 6.7 percentage points. By comparison, the linear and MLP routers contain 36 and 243 parameters. The experiment can consequently identify only large routing gains; smaller effects remain inconclusive. Because the features use raw teacher outputs, they may also encode confidence-scale differences in addition to content disagreement.

3.8 Budgeted Query Allocation

We compare random-batch and uncertainty-targeted allocation at the same realised teacher-query rate. All reported runs use the fixed teacher order

$$(T_1, T_2, T_3) = (\mathrm{Qwen}, \mathrm{Gemma}, \mathrm{Mistral}),$$

which is held constant across datasets, label budgets, and seeds. Every example receives T_1. Teachers are assigned equal unit query cost, so the budget q denotes the mean number of teacher queries per example.

Random-batch allocation. The equal-cost control samples a subset $\mathcal{R} \subseteq \mathcal{D}$ uniformly without replacement. Examples outside $\mathcal{R}$ receive only T_1, whereas examples in $\mathcal{R}$ receive all three teachers. For a requested query rate $q \in [1,3]$, the subset size is

$$m = \text{round}\left(\frac{N(q-1)}{2}\right), \qquad |\mathcal{R}| = m,$$

giving the realised query rate

$$\hat{q}_{\text{rand}} = 1 + \frac{2m}{N}.$$

The resulting consensus target is

$$\bar{p}^{\text{rand}}(x) = \begin{cases} \tilde{p}_1(x), & x \notin \mathcal{R}, \\ \dfrac{1}{3}\displaystyle\sum_{k=1}^{3} \tilde{p}_k(x), & x \in \mathcal{R}. \end{cases}$$

This distribution is converted to the distillation target using the same logarithm-and-temperature transformation as P_{casc}. The allocation seed is paired with the student-training seed, and the policy uses no gold labels.

This cost-matched control preserves the total query count but not the cascade's distribution of one-, two-, and three-teacher examples. A second control, defined below, additionally matches this query-depth profile.

Entropy-triggered cascade. The cascade first queries T_2 when the aligned predictive entropy of T_1 exceeds a threshold h. Among these escalated examples, it queries T_3 when T_1 and T_2 predict different classes. This extends uncertainty-based single-teacher escalation to the multi-teacher setting [20].

Let $\tilde{p}_k$ be the aligned distribution from Equation 3.12, and let $e_1(x) = H(\tilde{p}_1(x))$. The corresponding escalation sets are

$$\mathcal{S}_2(h) = \{x : e_1(x) > h\}, \qquad \mathcal{S}_3(h) = \left\{ x \in \mathcal{S}_2(h) : \arg\max_c \tilde{p}_{1,c}(x) \neq \arg\max_c \tilde{p}_{2,c}(x) \right\}. \tag{3.16}$$

The unsoftened cascade consensus is

$$\bar{p}^{\text{casc}}(x) = \begin{cases} \tilde{p}_1(x), & x \notin \mathcal{S}_2(h), \\ \frac{1}{2}\big(\tilde{p}_1(x) + \tilde{p}_2(x)\big), & x \in \mathcal{S}_2(h) \setminus \mathcal{S}_3(h), \\ \frac{1}{3}\sum_{k=1}^{3} \tilde{p}_k(x), & x \in \mathcal{S}_3(h). \end{cases} \tag{3.17}$$

It is softened using the same convention as the aligned mean:

$$P_{\text{casc}}(x) = \text{softmax}\left(\frac{\log \max\big(\bar{p}^{\text{casc}}(x), 10^{-12}\big)}{\tau}\right).$$

Budget calibration. The realised query rate is

$$\hat{q}(h) = 1 + \frac{|\mathcal{S}_2(h)|}{N} + \frac{|\mathcal{S}_3(h)|}{N}. \tag{3.18}$$

The threshold is selected to minimise $|\hat{q}(h) - q|$ for the requested rate q. Because $\hat{q}(h)$ is a monotone empirical step function, the requested and realised rates need not coincide exactly and both are reported.

Because T_3 is queried only when the first two teachers disagree, the cascade cannot reach $q = 3$ unless they disagree on every example. Its maximum query rate is

$$\sup_h \hat{q}(h) = 2 + d_{12}, \qquad d_{12} = \frac{1}{N}\left|\left\{x : \arg\max_c \tilde{p}_{1,c}(x) \neq \arg\max_c \tilde{p}_{2,c}(x)\right\}\right|. \tag{3.19}$$

The realised query rate is therefore bounded by $[1, 2 + d_{12}]$, with only a discrete set of rates attainable for a finite pool. Requests above $2 + d_{12}$ saturate at this upper bound. At the lower endpoint only T_1 is queried; at the upper endpoint T_2 is queried everywhere and T_3 only on disagreements. The full three-teacher aligned mean is not a member of this cascade family.

Depth-matched random allocation. This control preserves the cascade's query-depth profile while randomising which examples receive additional teachers. For the cascade sets at the selected threshold, let

$$n_1 = N - |\mathcal{S}_2|, \qquad n_2 = |\mathcal{S}_2 \setminus \mathcal{S}_3|, \qquad n_3 = |\mathcal{S}_3|.$$

A seeded random permutation partitions $\mathcal{D}$ into disjoint sets $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3$ of these respective sizes. Examples in $\mathcal{R}_j$ receive the first j teachers and use the corresponding branch of Equation 3.17. Consequently,

$$\hat{q}_{\text{depth}} = \frac{n_1 + 2n_2 + 3n_3}{N} = \hat{q}_{\text{casc}}.$$

The two policies therefore have the same teacher order and total query cost; only the selection of examples differs. The random allocation is paired with the training seed and uses no gold labels.

Offline-simulation qualification. The escalation decisions use only outputs that would be available online: T_1 entropy determines the second query, and disagreement between T_1 and T_2 determines the third. However, the principal experiment estimates all alignment parameters $\{b_k, t_k\}$ from the complete cache. An online cascade could estimate the parameters for T_2 and T_3 only from $\mathcal{S}_2$ and $\mathcal{S}_3$, respectively. A cached-logit sensitivity analysis, holding the escalation sets fixed, estimates the teacher-target effect of this optimistic assumption (Table A.12).

Evaluation. The cascade gives single-teacher targets to lower-entropy examples and averaged targets to escalated examples. Students are therefore trained from cascade, cost-matched random, and depth-matched random targets at matched $\hat{q}$, following Section 3.10. The first comparison evaluates complete allocation policies; the second isolates entropy-based example selection while holding query depth fixed. Target-level metrics are descriptive only.

3.9 Student Training

Every KD update matches a teacher-derived target, whereas gold-label cross-entropy is available only in minibatches containing labelled examples. All students use ModernBERT-base trained for three epochs with AdamW ($\eta = 2 \times 10^{-5}$, batch size 16, weight decay 0.01), with temperature $\tau = 2$ and loss coefficient $\alpha = 0.5$. These settings are fixed design choices rather than task-specific optima and were not selected using test performance. Across KD comparisons, only P_T changes; router runs additionally replace $\mathcal{D}_L$ with $\mathcal{D}_{L\text{-tr}}$. The complete configuration and procedure are reported in Table D.1 and Algorithm 5.

Objective and temperature convention. Let $\mathcal{D}_{\text{CE}} = \mathcal{D}_L$ in non-router runs and $\mathcal{D}_{\text{CE}} = \mathcal{D}_{L\text{-tr}}$ in router runs. For a KD minibatch $\mathcal{B} \subseteq \mathcal{D}$, let $\mathcal{B}_{\text{CE}} = \mathcal{B} \cap \mathcal{D}_{\text{CE}}$. The implemented objective is

$$\mathcal{L} = \alpha\tau^2 \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} D_{\text{KL}}\left(P_T[x] \parallel \text{softmax}(z_\theta(x)/\tau)\right) + (1-\alpha) \frac{1}{\max\{1, |\mathcal{B}_{\text{CE}}|\}} \sum_{(x,y) \in \mathcal{B}_{\text{CE}}} \mathcal{L}_{\text{CE}}(y, z_\theta(x)). \tag{3.20}$$

The cross-entropy sum is zero when $\mathcal{B}_{\text{CE}} = \varnothing$.

The KL term is implemented with log-softmax, and the standard τ^2 factor compensates for the reduced gradient scale introduced by temperature softening [14]. Targets P_T are softened once during construction, while the student distribution is softened in Equation 3.20; both use the same τ. Gold-label cross-entropy uses the unscaled student logits.

Sparse cross-entropy supervision. Cross-entropy is applied only in minibatches containing labelled examples and is averaged over the labelled members of those batches. Thus, α controls the relative coefficients when both loss terms are present, rather than

enforcing a training-wide 50:50 balance. The supervised baseline sets $\alpha = 0$ and trains directly on $\mathcal{D}_L$.

Validation and fixed configuration. Primary experiments reserve no validation set: at $L = 10$, even a 20% split on AG News would remove eight of the 40 labelled examples. They use no early stopping or checkpoint selection and are evaluated once after the fixed epoch budget. The supervised sensitivity analysis instead selects learning rate and epochs by five-fold cross-validation within $\mathcal{D}_L$, then retrains on all labels without using the test set (Section A.7).

Within the primary comparisons, architecture, optimiser, schedule, batch size, epoch budget, temperature, KD weight, working pool, and test set are fixed across methods.

3.10 Evaluation Protocol

3.10.1 Metrics and evaluation point

Students are evaluated once after the fixed training budget using deterministic arg-max test predictions; accuracy is primary and macro-F1 is descriptive. Test labels are never used for model selection. Harm AUC in Table 4.2 uses shuffled stratified five-fold out-of-fold predictions (seed 0) from L2-regularised logistic regression with unit inverse-regularisation strength, the `lbfgs` solver and a 1000-iteration limit, using unstandardised aggregate features 1–5 from Table B.1.

3.10.2 Paired comparisons and significance testing

Methods use the same ordered test examples under matched datasets, label budgets, seeds, and student configurations, making comparisons paired at the example level. Accuracy and macro-F1 are reported as mean $\pm$ sample standard deviation over three seeds for the primary comparisons, the $q = 2.0$ allocation experiment, and the depth-matched control. The original cost-matched comparison at $q = 1.4$ uses five seeds.

For each comparison and seed, an exact two-sided McNemar test [21] compares the numbers of test examples corrected and newly misclassified by one method relative to the other. Within each dataset–label-budget–comparison cell, p-values are Holm-adjusted [15] over the corresponding three or five seeds. Robustness requires a common direction across seeds and Holm-adjusted $p < 0.05$ for every seed; macro-F1 is descriptive.

Chapter 4

Results and Analysis

This chapter evaluates whether combining multiple teachers improves knowledge distillation under limited label and teacher-query budgets. Accuracy is reported as a percentage. Student results are the mean and sample standard deviation over seeds 42, 123 and 456 unless stated otherwise. Full accuracy and macro-F1 grids are reported in Appendix A, Tables A.1–A.4. Additional disagreement diagnostics and complete seed-level router results are reported in Tables A.5, A.6, and A.7.

4.1 Teacher Performance and Complementarity

Naive aggregation is inconsistent, whereas prior-and-scale alignment is the only aggregate in Table 4.1 that exceeds the best individual teacher on every training pool. The table establishes this comparison and reports the retrospective oracle opportunity available from teacher complementarity. The final row reports the proportion of examples for which at least one teacher predicts the correct class. Because it uses gold labels retrospectively, it measures complementarity rather than the performance of a deployable method.

Qwen is the strongest individual teacher on AG News, Yahoo and Amazon-3, leading the next-best teacher by 1.98, 0.60 and 4.26 percentage points, respectively. On Amazon, Gemma exceeds Qwen by only 0.04 points, equivalent to one example in the 2,500-example pool. Teacher ranking is therefore dataset-dependent, although Qwen is the strongest or effectively tied teacher across the four tasks.

Naive aggregation provides inconsistent benefits. Both the logit and probability means improve on the best individual teacher on Yahoo and Amazon, but are weaker on AG News and Amazon-3. Entropy weighting does not improve the logit mean on any

Table 4.1: Teacher and aggregate training-pool accuracy (%). Bold denotes the best non-oracle result; alignment is conditional on the supplied uniform prior.

Method	AG News	Yahoo	Amazon	Amazon-3
Individual teachers				
Qwen3.5-27B	88.18	64.90	62.76	79.13
Gemma-4-26B	86.13	64.30	62.80	74.87
Mistral-Small-24B	86.20	63.24	57.40	72.60
Aggregation rules				
Logit mean (B2)	87.05	65.08	63.84	76.07
Entropy-weighted logit mean (B4)	86.90	64.78	63.76	76.07
Probability mean (B3)	87.85	65.78	63.40	75.90
Prior-and-scale-aligned mean	**89.93**	**66.34**	**65.88**	**80.07**
Oracle: at least one teacher correct	91.03	71.74	77.44	85.93

dataset, consistent with the risk of assigning greater weight to teachers that are sharp but not reliably better calibrated. Probability averaging is stronger than logit averaging on AG News and Yahoo, but weaker on Amazon and Amazon-3; changing the aggregation space alone is therefore insufficient.

Among the aggregation rules in Table 4.1, prior-and-scale alignment is the only one that exceeds the best individual teacher on every dataset. Relative to probability averaging, it improves accuracy by 2.08, 0.56, 2.48 and 4.17 points on AG News, Yahoo, Amazon and Amazon-3, respectively. These gains show that class-marginal and confidence-scale mismatch account for a substantial part of the apparent teacher conflict.

The oracle opportunity rate remains 1.10, 5.40, 11.56 and 5.86 points above the aligned mean. This confirms that the teachers make complementary errors, particularly on Amazon, but it does not guarantee that a label-free aggregation rule can recover the gap. The more constrained class-conditional and soft Dawid–Skene diagnostics are reported in Table A.6; neither establishes a consistent advantage over the aligned mean.

4.2 Teacher Conflict and Target Alignment

Disagreement features identify harmful aggregate targets retrospectively, but disagreement volume alone does not determine how much harm aggregation causes. Table 4.2 reports both the prevalence of the harm event and how accurately it can be identified from the full-pool diagnostic features. Harm occurs when at least one teacher predicts

the correct class but the aggregate does not. Because both the harm labels and the AUC calculation use the gold training-pool labels, these results are retrospective diagnostics rather than deployable performance estimates.

Table 4.2: Training-pool disagreement and retrospective aggregation harm. Harm means that at least one teacher is correct but the aggregate is not; AUC is the stratified five-fold out-of-fold mean ± fold standard deviation.

Dataset	Top-1 disagree.	Mean sym. KL	Harm: logit	Harm: aligned	AUC: logit	AUC: aligned
AG News	6.56%	0.320	4.03%	2.72%	0.976 ± 0.002	0.950 ± 0.006
Yahoo	17.25%	0.950	6.70%	6.42%	0.926 ± 0.018	0.887 ± 0.007
Amazon	26.89%	1.109	13.64%	13.64%	0.843 ± 0.024	0.807 ± 0.019
Amazon-3	15.31%	0.666	9.87%	7.67%	0.955 ± 0.014	0.889 ± 0.026

AG News has both the lowest disagreement and the lowest harm rate, while Amazon has the highest values of both. The intermediate datasets show that the relationship is not one-to-one: Yahoo has more top-1 disagreement than Amazon-3 but a lower harm rate. Disagreement volume alone therefore does not determine how damaging aggregation will be.

The five aggregate features predict harm well when trained on labels from the full pool, with out-of-fold AUC between 0.843 and 0.976. Top-1 agreement is the strongest individual correlate on all four datasets, as reported in Table A.5. These results establish that harmful cases have a detectable retrospective structure, but do not show that the same structure can be learned reliably from the much smaller calibration sets available to the router.

Alignment reduces harm by 1.31 points on AG News, 0.28 on Yahoo and 2.20 on Amazon-3, but leaves the Amazon harm rate unchanged. On Amazon, alignment rescues 5.12% of the pool while newly harming another 5.12%. Its 2.04-point accuracy improvement over the logit mean must therefore arise from examples outside the harm definition, including cases where no individual teacher's argmax is correct but their corrected probabilities produce the correct aggregate prediction.

AUC decreases after alignment on every dataset. This is consistent with alignment removing confidence-scale effects that make some harmful cases easy to distinguish from raw outputs. However, this does not establish that all remaining errors reflect irreducible content disagreement: the same five raw-output features may simply be less effective at predicting harm after the target transformation.

Prior matching accounts for most of the improvement in target accuracy, whereas confidence-scale alignment primarily improves NLL and ECE. Table 4.3 separates these two corrections. The metrics are evaluated on the same transductive training pools used to fit them. Neither correction uses individual gold labels, although prior matching conditions on the supplied uniform class prior.

Table 4.3: Teacher-target accuracy/NLL/ECE on the balanced pools. Bold denotes the best value for each metric and dataset.

Aggregation	AG News	Yahoo	Amazon	Amazon-3
Probability mean (B3)	87.85/0.479/0.073	65.78/1.456/0.168	63.40/1.035/0.159	75.90/0.593/0.108
Scale alignment only	87.85/0.427/0.065	65.78/1.269/0.110	63.20/0.909/0.084	76.30/0.536/0.036
Prior matching only	**89.95**/0.376/0.046	**66.40**/1.286/0.153	64.92/0.922/0.144	**80.53**/0.501/0.069
Prior-and-scale alignment	89.93/**0.343/0.022**	66.34/**1.184/0.088**	**65.88/0.851/0.026**	80.07/**0.473/0.026**

Prior matching produces the highest accuracy on AG News, Yahoo and Amazon-3, while combined alignment is strongest on Amazon. Thus, most of the argmax improvement comes from correcting class-marginal bias rather than confidence scale. Scale alignment nevertheless improves NLL and ECE: the combined method obtains the lowest values of both metrics on all four datasets, even where its accuracy is slightly below prior matching alone.

This benefit depends on the supplied prior. Under strong misspecification, AG News target accuracy falls from 0.899 with the correct uniform prior to 0.650; the other three tasks show the same direction (Table A.11). The corresponding student-level sensitivity is evaluated in Section 4.3.1.

4.3 Student-Level Results

The principal student result is that correcting class-prior mismatch matters more reliably than adding teachers or aligning confidence scales. Table 4.4 compares the resulting students with both fixed and cross-validated supervised baselines and with raw Qwen as the pre-specified single-teacher reference. Results for every individual teacher are reported in Tables A.1–A.4. Probability averaging corresponds to the output-level component of You et al. [35]; their intermediate-representation constraints are not implemented because the LLM teachers are treated as black boxes.

The relationship between the evaluated baselines and published methods is sum-

Table 4.4: Student accuracy (%; mean ± standard deviation over three seeds). The fixed supervised baseline uses the common training configuration. The CV-selected baseline selects the learning rate and training duration by five-fold cross-validation within the labelled set. Raw Qwen is the fixed single-teacher baseline. Bold indicates the highest mean for each dataset and label budget.

Method	AG News	Yahoo	Amazon	Amazon-3
Panel A: $L = 10$				
Supervised (fixed)	65.09 ± 5.77	26.76 ± 2.66	22.00 ± 1.18	38.88 ± 1.67
Supervised (CV-selected)	72.00 ± 1.90	40.83 ± 4.50	26.05 ± 1.63	40.57 ± 0.46
Raw-Qwen KD	87.14 ± 0.49	59.20 ± 0.57	57.35 ± 0.82	72.11 ± 0.99
Logit-mean KD	85.70 ± 0.34	60.77 ± 0.43	58.19 ± 1.09	71.92 ± 1.42
Probability-mean KD	86.10 ± 0.41	60.48 ± 0.18	57.85 ± 1.75	72.24 ± 0.75
Prior-matched Qwen KD	$\mathbf{88.80 \pm 0.28}$	60.92 ± 0.21	$\mathbf{58.94 \pm 0.49}$	$\mathbf{74.06 \pm 1.17}$
Prior-matched committee KD	88.36 ± 0.23	$\mathbf{61.62 \pm 0.44}$	58.68 ± 0.67	73.90 ± 0.70
Prior-and-scale-aligned committee KD	88.47 ± 0.18	61.60 ± 0.13	58.18 ± 0.32	73.79 ± 0.47
Panel B: $L = 50$				
Supervised (fixed)	84.42 ± 1.30	52.14 ± 1.41	33.76 ± 2.35	47.66 ± 2.81
Supervised (CV-selected)	85.16 ± 1.60	54.50 ± 0.99	36.41 ± 2.53	57.64 ± 3.01
Raw-Qwen KD	87.08 ± 0.10	58.70 ± 1.45	57.32 ± 1.05	$\mathbf{73.76 \pm 0.24}$
Logit-mean KD	86.39 ± 1.28	60.55 ± 0.96	56.77 ± 0.67	72.33 ± 1.06
Probability-mean KD	86.66 ± 0.50	59.84 ± 1.10	56.45 ± 0.26	72.99 ± 0.72
Prior-matched Qwen KD	$\mathbf{88.81 \pm 0.07}$	60.55 ± 0.81	$\mathbf{57.33 \pm 1.23}$	73.24 ± 2.71
Prior-matched committee KD	88.46 ± 0.04	$\mathbf{61.32 \pm 0.34}$	57.24 ± 1.30	72.69 ± 1.29
Prior-and-scale-aligned committee KD	88.30 ± 0.32	60.57 ± 0.64	53.20 ± 1.51	72.84 ± 1.57

marised in Table A.13. In particular, the entropy-weighted baseline is an adapted heuristic inspired by Kwon et al. [18] and Cheng et al. [5], rather than an exact reproduction.

Raw-Qwen KD outperforms even the CV-selected supervised baseline in all eight comparisons. At $L = 10$, its advantages are 15.13, 18.38, 31.30, and 31.53 percentage points on AG News, Yahoo, Amazon, and Amazon-3, respectively. At $L = 50$, the corresponding advantages are 1.93, 4.20, 20.92, and 16.12 points. Seven comparisons satisfy the robustness criterion. On AG News at $L = 50$, the advantage is positive for all three seeds but does not satisfy the adjusted significance criterion. The complete comparison in Table A.14 shows a larger advantage at $L = 10$ on every task, supporting H2 within the two evaluated label budgets.

The main pattern is that correcting class-prior mismatch matters more than adding teachers. The prior-matched committee exceeds probability averaging in seven of eight

cells and does so robustly in five, indicating that uncorrected class-marginal bias is a major limitation of naive aggregation. However, the prior-matched Qwen student has the higher mean in six of eight direct comparisons with the committee. The committee leads only on Yahoo, and neither advantage satisfies the robustness criterion. Macro-F1 gives the same qualitative result.

One plausible explanation is the asymmetry of the teacher pool. Qwen is the strongest or effectively tied teacher across the four tasks, so adding teachers trades potential error complementarity against weaker or redundant signals. Yahoo provides some evidence that complementary teachers can help, but the uncertainty across seeds is too large to establish a reliable committee advantage. Multi-teacher aggregation is therefore useful relative to uncorrected baselines, but not consistently preferable to correcting the strongest teacher.

Confidence-scale alignment reveals a different limitation: a better teacher-level target is not necessarily easier for the student to learn. Combined alignment achieves the best teacher-level NLL and ECE, yet does not consistently improve student accuracy. At $L = 10$, the rankings of teacher-level and student-level gains have descriptive Spearman correlation $\rho = 0$. This separates target calibration from transfer: scale alignment preserves each teacher's predicted class but redistributes probability mass among the remaining classes, changing the soft-label structure presented to the student. Such changes may improve likelihood-based metrics without providing more useful supervision to a finite-capacity student, although the experiments do not isolate this mechanism.

Amazon at $L = 50$ provides the clearest example. Combined alignment reduces student accuracy from 57.24% for the prior-matched committee to 53.20%, despite producing a stronger teacher target with 65.88% accuracy and the lowest NLL. The reversal therefore arises during transfer rather than from a weaker teacher-level decision rule. The related Amazon-3 task does not show the same collapse, suggesting sensitivity to the five-class label formulation, although the present design cannot separate the effects of class count, ordinal ambiguity, and student optimisation.

Combined alignment also has the smallest seed standard deviation among the evaluated KD methods at $L = 10$, showing that stability across runs does not necessarily imply a better mean. These comparisons use the correct uniform prior; the next subsection tests sensitivity to that assumption.

4.3.1 Sensitivity to Prior Misspecification

Prior misspecification remains harmful after distillation. Mild misspecification lowers mean student accuracy in five of six conditions, while strong misspecification lowers all six by 8.19–24.27 percentage points. Table 4.5 tests this sensitivity using paired conditions in which the examples, labels, teacher logits, and student configurations are fixed and only the supplied prior changes.

Table 4.5: Student accuracy (%; mean $\pm$ standard deviation over three seeds) under correct, mildly misspecified and strongly misspecified priors. Parentheses give the paired mean change from the correct condition.

Dataset	L	Correct	Mild	Strong
AG News	10	88.47 ± 0.18	84.42 ± 1.12 (-4.05)	64.20 ± 5.17 (-24.27)
AG News	50	88.30 ± 0.32	84.48 ± 0.35 (-3.82)	67.89 ± 3.03 (-20.41)
Yahoo	10	61.60 ± 0.13	58.04 ± 0.75 (-3.56)	39.52 ± 3.02 (-22.08)
Yahoo	50	60.57 ± 0.64	58.97 ± 1.59 (-1.60)	47.98 ± 5.92 (-12.59)
Amazon	10	58.18 ± 0.32	54.27 ± 1.09 (-3.91)	35.51 ± 3.21 (-22.66)
Amazon	50	53.20 ± 1.51	53.24 ± 3.06 $(+0.04)$	45.01 ± 5.54 (-8.19)

Mild misspecification lowers mean accuracy in five of six cells and loses 16 of 18 seed-paired comparisons. The exception is Amazon at $L = 50$, where the $+0.04$-point mean difference masks mixed seed effects. Macro-F1 gives the same overall conclusion.

Strong misspecification lowers accuracy in all six cells and all 18 seed-paired comparisons; cell-level mean losses range from 8.19 to 24.27 points. The larger label budget attenuates the loss on Yahoo and Amazon but does not remove it. Thus, student training can attenuate mild prior error but does not protect against severe misspecification.

4.4 Learned-Router Feasibility

The learned router is not reliable in the evaluated low-label regime: the gate deploys it in only 3 of 18 runs, and none of those deployments improves on the same-seed logit mean. Table 4.6 reports the complete gate-controlled pipelines at $L = 10$. The reported student metrics evaluate the complete gated pipeline: when the calibration gate rejects the learned router, the student is trained from the selected entropy-weighted fallback.

They should therefore not be interpreted as learned-router accuracy in every seed.

Table 4.6: Gate-controlled router results at $L = 10$ (mean $\pm$ standard deviation over three seeds). Deployments count router selections; student metrics evaluate the resulting gated pipeline.

Dataset	Router	n_{cal}	Deployments	Accuracy	Macro-F1
AG News	Linear	12	$0/3$	86.34 ± 0.89	86.14 ± 0.94
AG News	MLP	12	$1/3$	86.17 ± 1.61	85.99 ± 1.67
Yahoo	Linear	30	$1/3$	60.07 ± 0.82	60.80 ± 0.45
Yahoo	MLP	30	$1/3$	60.05 ± 0.71	60.25 ± 0.92
Amazon	Linear	15	$0/3$	58.46 ± 0.69	57.26 ± 1.25
Amazon	MLP	15	$0/3$	58.06 ± 1.05	56.68 ± 1.49

The gate selects the entropy-weighted fallback in 15 of 18 runs. The learned router is deployed for the AG News MLP at seed 42 and for both Yahoo architectures at seed 456. In each case, its LOOCV advantage corresponds to exactly one additional correctly classified calibration example: $11/12$ rather than $10/12$ on AG News, and $19/30$ rather than $18/30$ on Yahoo.

The three deployed-router students score 85.04%, 59.64%, and 59.41%, compared with same-seed logit-mean accuracies of 85.59%, 60.28%, and 60.28%, respectively. The experiment therefore supports the safety gate as a conservative calibration-level filter, but not as a guarantee of improved student performance.

4.5 Budgeted Teacher Queries

Selective escalation does not improve the observed cost–accuracy frontier. Table 4.7 summarises the principal comparisons, while Appendix A retains the seed-level results. The $q = 1.0$ and $q = 3.0$ columns report retrospective envelopes: the strongest evaluated one-teacher target and the strongest evaluated full-committee target within each dataset–budget condition. This gives both alternatives a generous comparison, but neither envelope is itself a deployable model-selection policy. The cascade is a fixed allocation policy.

The strongest one-teacher target is prior-matched Qwen except on Amazon-3 at $L = 50$, where raw Qwen is stronger. The strongest full committee is the prior-matched committee in six conditions, the prior-and-scale-aligned committee on AG News at $L = 10$, and the probability mean on Amazon-3 at $L = 50$. Complete method-specific results are reported in Tables A.1–A.4.

Table 4.7: Cost-normalised student accuracy (%). The first three result columns report mean ± standard deviation over seeds 42, 123, and 456. The final columns report the difference between the cascade and cost-matched random allocation. Bold indicates the highest mean among the single-teacher, cascade, and full-committee results.

Dataset	L	Best single $q = 1.0$	Cascade $q = 1.4$	Best committee $q = 3.0$	$\Delta_{1.4}$	$\Delta_{2.0}$
AG News	10	**88.80 ± 0.28**	88.36 ± 0.15	88.47 ± 0.18	−0.28	−0.28
AG News	50	**88.81 ± 0.07**	88.46 ± 0.14	88.46 ± 0.04	−0.19	−0.03
Yahoo	10	60.92 ± 0.21	60.97 ± 0.54	**61.62 ± 0.44**	0.00	+0.48
Yahoo	50	60.55 ± 0.81	60.49 ± 0.66	**61.32 ± 0.34**	+0.13	−0.26
Amazon	10	**58.94 ± 0.49**	58.27 ± 0.48	58.68 ± 0.67	+0.26	−0.28
Amazon	50	**57.33 ± 1.23**	54.78 ± 1.21	57.24 ± 1.30	−0.14	−1.29
Amazon-3	10	**74.06 ± 1.17**	73.57 ± 0.83	73.90 ± 0.70	−0.20	−0.08
Amazon-3	50	**73.76 ± 0.24**	72.19 ± 1.45	72.99 ± 0.72	−0.01	−0.90

The final two columns compare the cascade with the original random-batch policy, which matches total cost but not query depth. An additional control randomises allocation while preserving the cascade's exact numbers of one-, two-, and three-teacher examples. This isolates entropy-based selection from query-depth effects. Complete results are reported in Tables A.8, A.9, and A.10.

Figure 4.1 brings together the target-to-student transfer analysis from Section 4.3 and the depth-matched allocation comparison.

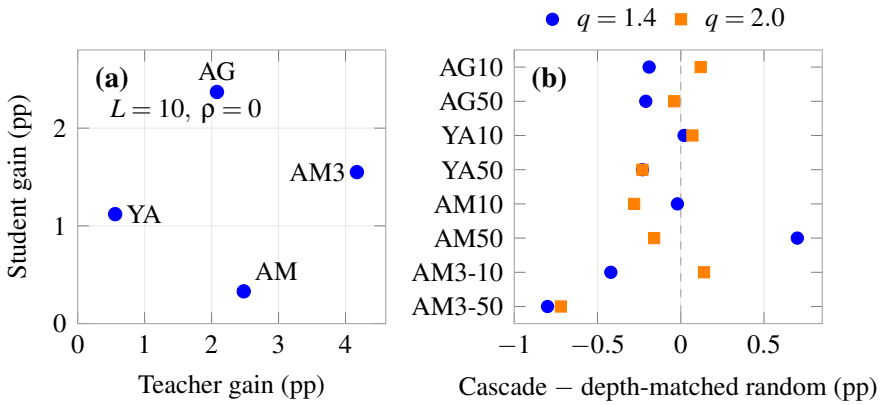


Figure 4.1: Panel (a) compares target- and student-level alignment gains over probability averaging at $L = 10$ (descriptive $\rho = 0$). Panel (b) shows no robust cascade advantage over depth-matched random allocation in 16 comparisons.

The strongest one-teacher target has the highest mean in six of eight conditions; the committee leads only on Yahoo, without a robust advantage. The cascade has no robust advantage over cost-matched random allocation at $q = 1.4$, while random allocation

has the higher mean in seven of eight conditions at $q = 2.0$, with one robust difference. None of the 16 depth-matched comparisons is robust (Table A.10). Entropy-based selection therefore does not improve the observed cost–accuracy frontier.

4.6 Answers to the Research Questions

Main research question. Conflict-aware purification improves naive aggregation, but the experiments do not establish an advantage over prior-matched Qwen or an improvement in the observed cost–accuracy frontier.

RQ1: diagnosing target harm and learning per-example corrections. Disagreement features identify harmful targets retrospectively, but the available calibration sets do not support a beneficial per-example router. The gate deploys the router in only 3 of 18 runs, none of which outperforms the same-seed logit mean.

RQ2: allocation under a fixed teacher-query budget. The cascade beats neither matched random control consistently; entropy does not reliably identify examples that benefit from additional teachers.

RQ3: transfer from targets to students. Teacher-target improvements do not transfer proportionally to students: their gain rankings have descriptive correlation $\rho = 0$ at $L = 10$, and Amazon exhibits a transfer reversal at $L = 50$. Target metrics therefore cannot replace trained-student evaluation.

Hypotheses. H1 and H3 are not supported: across the four tasks, neither target-level nor student-level gains from prior-and-scale alignment over probability averaging increase consistently with top-1 disagreement (Tables 4.2–4.4), and the cascade does not consistently beat either random control. H2 is supported within the evaluated budgets because raw-Qwen KD has a larger advantage over supervised training at $L = 10$ than at $L = 50$ on every task.

Chapter 5

Discussion and Limitations

5.1 From Teacher Conflict to Calibration

Multi-teacher distillation is often framed as deciding which teacher to trust when predictions conflict [16, 44]. The results show that raw disagreement also includes systematic class preferences and differences in confidence scale. These effects matter because uniform logit averaging produces a normalised geometric mean of the teachers' probability distributions before the KD temperature is applied: a teacher with greater logit spread can exert more influence despite nominally equal weights.

The alignment ablation separates the two corrections. Prior matching can change top-1 decisions and produces the highest target accuracy on three tasks. Confidence-scale alignment is order-preserving for each teacher and primarily improves NLL and ECE. Prior matching therefore corrects systematic class preference, whereas scale alignment modifies the confidence structure of the target.

Alignment also improves cases outside the stated harm definition. On Amazon, relative to the raw logit mean, prior-and-scale alignment leaves the harm rate unchanged at 13.64% but raises target accuracy from 63.84% to 65.88%, an increase of 2.04 percentage points. The net accuracy gain must therefore come from cases outside the harm event, including examples for which no individual teacher has the correct argmax but the aligned aggregate predicts the correct class.

Prior matching is beneficial only when the supplied prior is reasonably accurate. The target and student analyses agree that severe prior errors survive distillation, while the effects of mild misspecification are smaller and less uniform. A larger labelled budget attenuates some effects but does not eliminate strong-prior failure. The exact uniform prior available in the principal experiments is therefore a substantive condition

of the reported benefit.

5.2 Why Better Targets Do Not Guarantee Better Students

The transfer results suggest that better calibrated targets are not necessarily more teachable soft labels. Amazon receives the strongest confidence-scale correction: the fitted temperatures 2.024, 4.802, and 1.000 raise mean target entropy from 0.563 after prior matching to 0.868 nats after scale alignment. The KD temperature $\tau = 2$ raises it further to 1.309 (Table A.15). This is consistent with over-softening, but it does not establish over-softening as the cause because temperature, KD weight, training duration, and student architecture are not varied.

More generally, closer teacher matching need not improve student generalisation [30], while calibration and the teacher-student gap can also affect distillability [37]. Teacher-target metrics are therefore useful diagnostics, but their practical value must be established through a trained student.

5.3 When Per-Sample Routing May Be Useful

The completed experiment does not support per-sample routing in the evaluated regime: the gate selects the router in only 3 of 18 runs, each by a one-example LOOCV margin, and none outperforms the same-seed naive logit mean. Calibration accuracy is therefore not a reliable proxy for downstream gain.

This instability is consistent with fitting 36- or 243-parameter routers from only 12–30 calibration examples. Oracle diagnostics show that complementarity exists, but exploiting it may require more calibration data, lower-capacity rules, aligned features, or a more heterogeneous teacher pool.

5.4 Why Selective Query Allocation Did Not Pay

The entropy cascade provides no consistent advantage over either random control. At the larger budget it approaches its attainable ceiling and queries the second teacher on most examples, leaving little scope for selectivity. Entropy also misses confident errors, while uncertain examples need not benefit when teachers make similar mistakes.

The depth-matched control preserves the cascade's one-, two-, and three-teacher counts, yet none of its 16 comparisons is robust. The original depth-profile difference therefore does not explain the absence of a cascade advantage. This conclusion remains specific to one fixed teacher order and equal unit query costs.

5.5 Implications for Multi-Teacher Distillation

The results support a staged workflow. Practitioners should first diagnose systematic output differences, then compare the committee with pre-specified corrected single-teacher baselines through trained students. Routing or selective querying should be introduced only when sufficient residual complementarity remains.

This qualifies earlier findings that enlarging a teacher pool can reduce student performance [16]. In the present setting, part of the damage arises from correctable output mismatch, but correcting that mismatch does not guarantee that a committee will outperform one corrected teacher.

5.6 Limitations

Prior matching assumes a supplied class prior, which is exactly correct for the balanced principal experiments. Student-level sensitivity is evaluated on three datasets under mild and strong perturbations, but only for one predeclared dominant class. The experiment also retains balanced pools and supplies the perturbed prior directly; it does not test naturally unbalanced data, deployment-time prior estimation, covariate shift, or transfer of alignment parameters between batches. Alignment remains transductive, and the cascade subset analysis does not retrain students using subset-estimated parameters.

The pool contains three similarly sized teachers and only three independent English tasks because the Amazon variants share their source data. Results may differ with heterogeneous teachers, tasks, or costs. Moreover, the public benchmarks predate the teachers and their pretraining corpora are not fully disclosed, so contamination cannot be excluded; the findings concern distillation on these benchmarks rather than teacher generalisation to unseen data.

Primary KD comparisons use a single student architecture and fixed optimisation settings. This is relevant because encoder fine-tuning can be sensitive to hyperparameters and random seeds [24]. The supervised CV sweep selects the longest schedule in all eight cells and the largest learning rate in seven, suggesting that the optimum may lie

beyond the tested grid. It therefore checks sensitivity rather than establishing the best possible supervised result. Primary comparisons and the depth-matched control use three seeds, while the original $q = 1.4$ cost-matched comparison uses five. Small mean differences should therefore be interpreted cautiously. Alternative KD objectives and logit normalisation can change how teacher information is transferred [41, 31]; whether these choices, another temperature, or a different student architecture would remove the Amazon reversal remains open.

Cost and access model. Teachers are assigned equal unit query cost, which isolates the allocation decision but does not reflect deployments in which models differ in price, latency, or availability. Inference-time systems such as FrugalGPT and RouteLLM derive part of their value from routing between models with heterogeneous capabilities and costs [4, 25]. Such asymmetry could change the preferred teacher order and make cascading economically useful even when it does not improve accuracy at equal query count. The present experiments therefore support conclusions about query allocation, not realised monetary or latency savings.

Teacher interface and prompting. Teachers are accessed only through constrained-decoding class-token log-probabilities. This interface is inexpensive and permits controlled aggregation, but excludes explanatory information available through generated answers or rationales [32, 16]. The prompt template, demonstration order, and verbalizer assignment are also fixed. Prompted classifiers can exhibit systematic sensitivity to these choices [42], so part of the class-marginal mismatch corrected by prior matching may originate in the chosen interface. Whether prompt-level calibration and output-level alignment are substitutes or complementary interventions is not established.

Finally, routing is evaluated only at $L = 10$, with three seeds, 12–30 calibration examples, and features computed from unaligned teacher outputs. The allocation experiments use one teacher order and equal unit costs. Together, these limitations and the output-level adaptations of literature baselines restrict the conclusions to the evaluated pipeline rather than multi-teacher distillation generally.

Chapter 6

Conclusions and Future Work

6.1 Summary of Findings

This dissertation examined whether conflict-aware purification can improve multi-teacher knowledge distillation under limited label and teacher-query budgets. Three findings emerge. First, disagreement features identify harmful aggregation retrospectively, but the available calibration sets do not support reliable per-example routing. Second, prior matching improves naive aggregation when the supplied prior is accurate, but the strongest single-teacher result has the highest mean in six of eight cost-normalised conditions, and teacher-target improvements do not otherwise transfer consistently to students. Third, entropy-based cascading does not outperform random query allocation at matched cost or query depth.

H1 is not supported because purification gains do not increase consistently with teacher disagreement. H2 is supported within the two evaluated label budgets because the advantage of raw-Qwen KD over the CV-selected supervised baseline is larger at $L = 10$ than at $L = 50$ on every task. H3 is not supported because the cascade provides no consistent advantage over either random control.

6.2 Conclusions

The main research question has a conditional answer. Conflict-aware purification can improve naive multi-teacher distillation, but the most reliable intervention in these experiments is pool-level calibration rather than per-example routing. When the balanced-pool prior is supplied correctly, correcting class-marginal bias improves every evaluated teacher target and improves students over probability averaging in seven

of eight dataset–budget conditions, without requiring additional teacher queries. Strong misspecification reduces student accuracy by 8.19–24.27 points across all six tested cells, making reliable prior information a necessary condition for that benefit. This finding extends prior-adjustment and contextual-calibration ideas [28, 42] to the multi-teacher aggregation setting. Confidence-scale alignment improves target calibration, but its student-level value depends on the task and training configuration.

The experiments do not establish a general advantage for querying multiple teachers. The committee has a higher mean than prior-matched Qwen only on Yahoo, but neither difference is robust. Additional teachers may help when their complementary information can be recovered without diluting a strong individual teacher. This result qualifies earlier observations that enlarging a teacher pool can reduce student performance [16]: in the present setting, part of the degradation arises from correctable output mismatch rather than irreducible knowledge conflict. For the comparatively homogeneous pool studied here, the evaluated prior-matched Qwen baseline is often as effective as aggregating all three; whether an equally strong teacher can be identified prospectively remains unresolved.

The broader methodological conclusion is that teacher-target quality and student teachability are distinct. This complements distillation methods that treat target geometry and logit scale as important independently of top-1 teacher correctness [41, 31]. Accuracy, NLL, and ECE are valuable diagnostics, but cannot replace evaluation through a trained student. Similarly, target-level allocation simulations cannot substitute for student-level comparisons at matched cost and query depth.

Multi-teacher distillation should therefore proceed incrementally: diagnose teacher outputs, correct systematic calibration differences, validate a corrected single-teacher baseline, and add routing or selective querying only when sufficient residual complementarity remains.

6.3 Future Work

The first priority is to evaluate prior estimation under deployment-like conditions. Future work should estimate the prior from a separate unlabelled batch, test naturally imbalanced pools, and determine whether corrections fitted on one batch transfer to another.

The second priority is to identify the cause of the confidence-scale transfer reversal. The fitted scales and entropy diagnostics make over-softening plausible, but experiments

varying distillation temperature, KD weight, training duration, and student architecture are required to distinguish a target defect from an optimisation interaction.

Routing should be revisited with a larger calibration set, lower-capacity decision rules, and features computed from aligned teacher distributions. A more heterogeneous pool containing different model families, parameter scales, or domain-specialised teachers would provide a stronger test of whether per-example expertise can be learned.

Finally, selective query allocation should be tested with alternative teacher orders and heterogeneous query costs. Future work could adapt uncertainty-based distillation allocation [20] and cost-aware inference routing [4, 25] to the student-training setting. A prospective implementation should estimate each teacher's alignment parameters only from queried examples and could escalate on predicted marginal improvement rather than first-teacher entropy alone.

Together, these extensions would establish whether the positive calibration result generalises beyond balanced offline pools and whether routing and selective querying become valuable when teacher complementarity is stronger and the available supervision is sufficient to identify it.

Bibliography

[1] Cenk Baykal, Khoa Trinh, Fotis Iliopoulos, Gaurav Menghani, and Erik Vee. Robust active distillation. In *Proceedings of the 11th International Conference on Learning Representations, ICLR '23*, 2023.

[2] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems 33, NeurIPS '20*, pages 1877–1901, 2020.

[3] Lianshang Cai, Linhao Zhang, Dehong Ma, Jun Fan, Daiting Shi, Yi Wu, Zhicong Cheng, Simiu Gu, and Dawei Yin. PILE: Pairwise iterative logits ensemble for multi-teacher labeled distillation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 587–595, 2022.

[4] Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *Transactions on Machine Learning Research*, 2024.

[5] Xin Cheng, Jialiang Tang, Zhiqiang Zhang, Wenxin Yu, Ning Jiang, and Jinjia Zhou. Decoupled multi-teacher knowledge distillation based on entropy. In *2024 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 2174–2178, 2024.

[6] Jang Hyun Cho and Bharath Hariharan. On the efficacy of knowledge distillation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4794–4802, 2019.

[7] Alexander P. Dawid and Allan M. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(1):20–28, 1979.

[8] Shangchen Du, Shan You, Xiaojie Li, Jianlong Wu, Fei Wang, Chen Qian, and Changshui Zhang. Agree to disagree: Adaptive ensemble knowledge distillation in gradient space. In *Advances in Neural Information Processing Systems*, volume 33, pages 12345–12355, 2020.

[9] Wen-Shu Fan, Su Lu, Xin-Chun Li, De-Chuan Zhan, and Le Gan. Revisit the essence of distilling knowledge through calibration. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 12882–12894, 2024.

[10] Gemma Team. Gemma 4 technical report. arXiv preprint arXiv:2607.02770, 2026.

[11] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proc. of the 34th Intl. Conf. on Machine Learning, ICML '17*, pages 1321–1330, 2017.

[12] Faisal Hamman, Pasan Dissanayake, Yanjun Fu, and Sanghamitra Dutta. Few-shot knowledge distillation of LLMs with counterfactual explanations. *arXiv preprint arXiv:2510.21631*, 2025.

[13] Tom Heskes. Selecting weighting factors in logarithmic opinion pools. In *Advances in Neural Information Processing Systems*, volume 10, pages 266–272, 1997.

[14] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[15] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[16] Ruihan Jin, Pengpeng Shao, Zhengqi Wen, Jinyang Wu, Mingkuan Feng, Shuo Yang, Chu Yuan Zhang, and Jianhua Tao. Exploring knowledge purification in multi-teacher knowledge distillation for LLMs. In *Proceedings of the Fourteenth International Conference on Learning Representations*, 2026.

[17] M. G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1–2):81–93, 1938.

[18] Kisoo Kwon, Hwidong Na, Hoshik Lee, and Nam Soo Kim. Adaptive knowledge distillation based on entropy. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP '20*, pages 7409–7413, 2020.

[19] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626, 2023.

[20] Viviana Luccioli, Rithika Iyengar, Ryan Panley, Flora Haberkorn, Xiaoyu Ge, Leland Crane, Nitish Sinha, and Seung Jung Lee. LLM on a budget: Active knowledge distillation for efficient classification of large text corpora. Finance and Economics Discussion Series 2025-108, Board of Governors of the Federal Reserve System, 2025.

[21] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.

[22] Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. Long-tail learning via logit adjustment. In *Proc. of the 9th Intl. Conf. on Learning Representations, ICLR '21*, 2021.

[23] Mistral AI. Mistral-Small-3.2-24B-Instruct-2506. Hugging Face model card, `https://huggingface.co/mistralai/Mistral-Small-3.2-24B-Instruct-2506`, June 2025. Accessed: 2026-07-29.

[24] Marius Mosbach, Maksym Andriushchenko, and Dietrich Klakow. On the stability of fine-tuning BERT: Misconceptions, explanations, and strong baselines. In *International Conference on Learning Representations*, 2021.

[25] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs from preference data. In *Proceedings of the Thirteenth International Conference on Learning Representations*, 2025.

[26] Flavio Di Palo, Prateek Singhi, and Bilal H. Fadlallah. Performance-guided LLM knowledge distillation for efficient text classification at scale. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 3675–3687, 2024.

[27] Qwen Team. Qwen3.5-27B. Hugging Face model card, `https://huggingface.co/Qwen/Qwen3.5-27B`, 2026. Accessed: 2026-08-18.

[28] Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002.

[29] Gaurav Sahu, Olga Vechtomova, Dzmitry Bahdanau, and Issam Laradji. PromptMix: A class boundary augmentation method for large language model distillation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5316–5327, 2023.

[30] Samuel Stanton, Pavel Izmailov, Polina Kirichenko, Alexander A. Alemi, and Andrew Gordon Wilson. Does knowledge distillation really work? In *Advances in Neural Information Processing Systems*, volume 34, pages 6906–6919, 2021.

[31] Shangquan Sun, Wenqi Ren, Jingzhi Li, Rui Wang, and Xiaochun Cao. Logit standardization in knowledge distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR '24*, pages 15731–15740, 2024.

[32] Yijun Tian, Yikun Han, Xiusi Chen, Wei Wang, and Nitesh V. Chawla. Beyond answers: Transferring reasoning capabilities to smaller LLMs using multi-teacher knowledge distillation. In *Proceedings of the 18th ACM International Conference on Web Search and Data Mining, WSDM '25*, pages 251–260, 2025.

[33] Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2526–2547, 2025.

[34] Danny Wood, Tingting Mu, Andrew M. Webb, Henry W. J. Reeve, Mikel Luján, and Gavin Brown. A unified theory of diversity in ensemble learning. *Journal of Machine Learning Research*, 24(359):1–49, 2023.

[35] Shan You, Chang Xu, Chao Xu, and Dacheng Tao. Learning from multiple teacher networks. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '17*, pages 1285–1294, 2017.

[36] Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, Ming Gong, Yan Fu, and Daxin Jiang. Reinforced multi-teacher selection for knowledge distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 14284–14291, 2021.

[37] Weixiao Zhan, Yongcheng Jing, Leszek Rutkowski, and Dacheng Tao. Distillation traps and guards: A calibration knob for LLM distillability. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 19814–19833, 2026.

[38] Hailin Zhang, Defang Chen, and Can Wang. Confidence-aware multi-teacher knowledge distillation. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP '22*, pages 4498–4502, 2022.

[39] Hengyuan Zhang, Shiping Yang, Xiao Liang, Chenming Shang, Yuxuan Jiang, Chaofan Tao, Jing Xiong, Hayden Kwok-Hay So, Ruobing Xie, Angel X. Chang, and Ngai Wong. Find your optimal teacher: Personalized data synthesis via router-guided multi-teacher distillation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 14619–14637, 2026.

[40] Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems 28, NeurIPS '15*, pages 649–657, 2015.

[41] Borui Zhao, Quan Cui, Renjie Song, Yiyu Qiu, and Jiajun Liang. Decoupled knowledge distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR '22*, pages 11953–11962, 2022.

[42] Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. In *Proc. of the 38th Intl. Conf. on Machine Learning, ICML '21*, pages 12697–12706, 2021.

[43] Han Zhou, Xingchen Wan, Lev Proleev, Diana Mincu, Jilin Chen, Katherine A. Heller, and Subhrajit Roy. Batch calibration: Rethinking calibration for in-context learning and prompt engineering. In *International Conference on Learning Representations*, 2024.

[44] Wenjie Zhou, Zhenxin Ding, Xiaodong Zhang, Haibo Shi, Junfeng Wang, and Dawei Yin. GOVERN: Gradient orientation vote ensemble for multi-teacher

reinforced distillation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1650–1658, 2024.

Appendix A

Supplementary Results

This appendix reports the complete student grids and secondary diagnostics supporting Chapter 4. Unless stated otherwise, student results are percentages reported as the mean and sample standard deviation over seeds 42, 123 and 456.

A.1 Complete Student Results

Tables A.1–A.4 report accuracy and macro-F1 for every evaluated single-teacher, aggregation and alignment method. Bold indicates the highest mean accuracy within each label budget; the corresponding macro-F1 is bolded in the same cell.

Table A.1: Complete AG News student results. Values are accuracy/macro-F1, reported as mean ± seed standard deviation.

Method	$L = 10$	$L = 50$
Supervised	$65.09 \pm 5.77 \,/\, 64.13 \pm 5.05$	$84.42 \pm 1.30 \,/\, 84.32 \pm 1.34$
KD ← raw Qwen	$87.14 \pm 0.49 \,/\, 87.02 \pm 0.55$	$87.08 \pm 0.10 \,/\, 86.96 \pm 0.10$
KD ← raw Gemma	$84.80 \pm 0.41 \,/\, 84.43 \pm 0.48$	$85.35 \pm 2.25 \,/\, 84.99 \pm 2.58$
KD ← raw Mistral	$86.42 \pm 0.38 \,/\, 86.29 \pm 0.42$	$86.58 \pm 0.50 \,/\, 86.45 \pm 0.52$
KD ← prior-matched Qwen	$\mathbf{88.80 \pm 0.28} \,/\, \mathbf{88.78 \pm 0.28}$	$\mathbf{88.81 \pm 0.07} \,/\, \mathbf{88.79 \pm 0.06}$
KD ← logit mean	$85.70 \pm 0.34 \,/\, 85.44 \pm 0.35$	$86.39 \pm 1.28 \,/\, 86.16 \pm 1.42$
KD ← entropy-weighted logit mean	$85.80 \pm 0.38 \,/\, 85.53 \pm 0.40$	$86.50 \pm 0.71 \,/\, 86.30 \pm 0.79$
KD ← probability mean	$86.10 \pm 0.41 \,/\, 85.89 \pm 0.44$	$86.66 \pm 0.50 \,/\, 86.50 \pm 0.54$
KD ← prior-matched mean	$88.36 \pm 0.23 \,/\, 88.32 \pm 0.24$	$88.46 \pm 0.04 \,/\, 88.43 \pm 0.05$
KD ← scale-aligned mean	$86.22 \pm 0.41 \,/\, 86.02 \pm 0.46$	$86.79 \pm 0.42 \,/\, 86.65 \pm 0.47$
KD ← prior-and-scale-aligned mean	$88.47 \pm 0.18 \,/\, 88.44 \pm 0.19$	$88.30 \pm 0.32 \,/\, 88.27 \pm 0.33$

Table A.2: Complete Yahoo student results. Values are accuracy/macro-F1, reported as mean ± seed standard deviation.

Method	$L = 10$	$L = 50$
Supervised	$26.76 \pm 2.66 \,/\, 24.92 \pm 2.19$	$52.14 \pm 1.41 \,/\, 52.30 \pm 1.33$
KD ← raw Qwen	$59.20 \pm 0.57 \,/\, 59.98 \pm 0.39$	$58.70 \pm 1.45 \,/\, 59.05 \pm 1.25$
KD ← raw Gemma	$60.30 \pm 0.25 \,/\, 60.65 \pm 0.36$	$59.60 \pm 1.50 \,/\, 59.82 \pm 1.43$
KD ← raw Mistral	$59.78 \pm 0.41 \,/\, 59.46 \pm 0.62$	$60.23 \pm 0.93 \,/\, 59.71 \pm 0.73$
KD ← prior-matched Qwen	$60.92 \pm 0.21 \,/\, 60.84 \pm 0.24$	$60.55 \pm 0.81 \,/\, 60.58 \pm 0.72$
KD ← logit mean	$60.77 \pm 0.43 \,/\, 61.02 \pm 0.40$	$60.55 \pm 0.96 \,/\, 60.56 \pm 0.93$
KD ← entropy-weighted logit mean	$60.45 \pm 1.07 \,/\, 60.83 \pm 0.97$	$60.06 \pm 0.66 \,/\, 60.46 \pm 0.60$
KD ← probability mean	$60.48 \pm 0.18 \,/\, 60.73 \pm 0.26$	$59.84 \pm 1.10 \,/\, 59.79 \pm 0.77$
KD ← prior-matched mean	$\mathbf{61.62 \pm 0.44 \,/\, 61.50 \pm 0.38}$	$\mathbf{61.32 \pm 0.34 \,/\, 61.24 \pm 0.36}$
KD ← scale-aligned mean	$60.00 \pm 0.14 \,/\, 60.13 \pm 0.23$	$60.37 \pm 0.94 \,/\, 60.23 \pm 0.60$
KD ← prior-and-scale-aligned mean	$61.60 \pm 0.13 \,/\, 61.40 \pm 0.19$	$60.57 \pm 0.64 \,/\, 60.50 \pm 0.66$

Table A.3: Complete Amazon student results. Values are accuracy/macro-F1, reported as mean ± seed standard deviation.

Method	$L = 10$	$L = 50$
Supervised	$22.00 \pm 1.18 \,/\, 16.50 \pm 3.47$	$33.76 \pm 2.35 \,/\, 30.38 \pm 1.21$
KD ← raw Qwen	$57.35 \pm 0.82 \,/\, 54.86 \pm 1.94$	$57.32 \pm 1.05 \,/\, 55.26 \pm 2.80$
KD ← raw Gemma	$57.15 \pm 0.74 \,/\, 55.59 \pm 1.37$	$55.81 \pm 0.92 \,/\, 54.28 \pm 0.85$
KD ← raw Mistral	$52.41 \pm 2.80 \,/\, 47.79 \pm 5.12$	$53.73 \pm 0.58 \,/\, 51.10 \pm 1.67$
KD ← prior-matched Qwen	$\mathbf{58.94 \pm 0.49 \,/\, 58.75 \pm 0.69}$	$\mathbf{57.33 \pm 1.23 \,/\, 57.36 \pm 0.81}$
KD ← logit mean	$58.19 \pm 1.09 \,/\, 56.60 \pm 1.72$	$56.77 \pm 0.67 \,/\, 55.48 \pm 0.86$
KD ← entropy-weighted logit mean	$58.49 \pm 0.46 \,/\, 57.56 \pm 1.16$	$56.20 \pm 1.21 \,/\, 54.57 \pm 2.23$
KD ← probability mean	$57.85 \pm 1.75 \,/\, 56.10 \pm 2.67$	$56.45 \pm 0.26 \,/\, 55.13 \pm 1.04$
KD ← prior-matched mean	$58.68 \pm 0.67 \,/\, 58.17 \pm 1.26$	$57.24 \pm 1.30 \,/\, 57.02 \pm 1.11$
KD ← scale-aligned mean	$56.70 \pm 0.78 \,/\, 54.54 \pm 1.27$	$56.06 \pm 0.11 \,/\, 54.70 \pm 0.72$
KD ← prior-and-scale-aligned mean	$58.18 \pm 0.32 \,/\, 58.02 \pm 0.61$	$53.20 \pm 1.51 \,/\, 52.27 \pm 1.44$

Table A.4: Complete Amazon-3 student results. Values are accuracy/macro-F1, reported as mean ± seed standard deviation.

Method	$L = 10$	$L = 50$
Supervised	$38.88 \pm 1.67 \,/\, 37.46 \pm 1.69$	$47.66 \pm 2.81 \,/\, 46.40 \pm 3.78$
KD ← raw Qwen	$72.11 \pm 0.99 \,/\, 70.30 \pm 1.34$	$\mathbf{73.76 \pm 0.24} \,/\, \mathbf{73.58 \pm 0.24}$
KD ← raw Gemma	$69.43 \pm 2.88 \,/\, 66.03 \pm 4.01$	$68.81 \pm 2.87 \,/\, 66.49 \pm 3.16$
KD ← raw Mistral	$67.97 \pm 0.68 \,/\, 63.68 \pm 1.82$	$72.36 \pm 0.22 \,/\, 71.31 \pm 0.17$
KD ← prior-matched Qwen	$\mathbf{74.06 \pm 1.17} \,/\, \mathbf{73.53 \pm 1.84}$	$73.24 \pm 2.71 \,/\, 73.28 \pm 2.53$
KD ← logit mean	$71.92 \pm 1.42 \,/\, 69.74 \pm 2.50$	$72.33 \pm 1.06 \,/\, 71.26 \pm 1.23$
KD ← entropy-weighted logit mean	$70.37 \pm 1.70 \,/\, 67.23 \pm 3.01$	$72.37 \pm 1.07 \,/\, 71.25 \pm 0.98$
KD ← probability mean	$72.24 \pm 0.75 \,/\, 70.17 \pm 1.41$	$72.99 \pm 0.72 \,/\, 72.12 \pm 1.06$
KD ← prior-matched mean	$73.90 \pm 0.70 \,/\, 73.28 \pm 1.24$	$72.69 \pm 1.29 \,/\, 72.63 \pm 1.37$
KD ← scale-aligned mean	$70.41 \pm 1.55 \,/\, 67.68 \pm 2.45$	$72.72 \pm 0.76 \,/\, 71.60 \pm 1.32$
KD ← prior-and-scale-aligned mean	$73.79 \pm 0.47 \,/\, 73.48 \pm 0.86$	$72.84 \pm 1.57 \,/\, 72.93 \pm 1.52$

A.2 Additional Conflict Diagnostics

Table A.5: Point-biserial correlation between disagreement features and the retrospective harm indicator. Teacher-specific divergences are ordered as Qwen/Gemma/Mistral.

Feature	AG News	Yahoo	Amazon	Amazon-3
Consensus entropy	0.55	0.36	0.36	0.48
Mean pairwise KL	0.57	0.40	0.37	0.47
Maximum pairwise KL	0.55	0.36	0.35	0.44
Top-1 agreement	−0.63	−0.49	−0.50	−0.61
Mean Kendall's τ	0.03	0.06	−0.12	−0.30
Top-3 class variance	0.59/0.59/0.16	0.41/0.44/0.28	0.44/0.42/0.17	0.58/0.55/0.23
Teacher KL from consensus	0.44/0.58/0.47	0.35/0.44/0.34	0.37/0.41/0.22	0.47/0.57/0.22

Table A.6: Class-conditional routing diagnostics on the training pools. Oracle and supervised rows use gold training-pool labels. The soft Dawid–Skene-inspired posterior replaces hard votes with prior-and-scale-aligned teacher probabilities. It is initialised from their mean and fitted for 50 EM iterations using a uniform latent-class prior and unit pseudocount smoothing. Values are percentages.

Method	AG News	Yahoo	Amazon
Prior-and-scale-aligned mean	89.93	66.34	65.88
Oracle best teacher for each true class	90.40	67.26	67.52
Oracle convex weights for each true class	90.87	68.18	68.48
Supervised expertise weights at predicted class	89.93	66.34	65.96
Soft Dawid–Skene-inspired posterior	89.82	66.08	65.04

A.3 Complete Learned-Router Results

Table A.7 reports every gate decision in the declared router experiment. The fixed calibration accuracy is the stronger of the uniform-logit and entropy-weighted heuristics; the two heuristics tie in these runs. Student accuracy and macro-F1 evaluate the resulting gated pipeline.

Table A.7: Seed-level router results at $L = 10$. a_r is the selected router's LOOCV accuracy and a_f is the best fixed-rule calibration accuracy. The gate deploys the router only when $a_r > a_f$. All values are percentages.

Dataset	Router	Seed	n_{cal}	a_r	a_f	Gate	Accuracy	Macro-F1
AG News	Linear	42	12	83.3	83.3	Entropy	86.58	86.38
AG News	MLP	42	12	91.7	83.3	Router	85.04	84.86
AG News	Linear	123	12	75.0	75.0	Entropy	85.36	85.10
AG News	MLP	123	12	75.0	75.0	Entropy	85.45	85.20
AG News	Linear	456	12	83.3	83.3	Entropy	87.09	86.93
AG News	MLP	456	12	83.3	83.3	Entropy	88.01	87.91
Yahoo	Linear	42	30	56.7	56.7	Entropy	59.56	60.42
Yahoo	MLP	42	30	56.7	56.7	Entropy	59.94	60.70
Yahoo	Linear	123	30	76.7	76.7	Entropy	61.01	61.29
Yahoo	MLP	123	30	76.7	76.7	Entropy	60.81	60.86
Yahoo	Linear	456	30	63.3	60.0	Router	59.64	60.69
Yahoo	MLP	456	30	63.3	60.0	Router	59.41	59.18
Amazon	Linear	42	15	73.3	73.3	Entropy	57.69	55.96
Amazon	MLP	42	15	73.3	73.3	Entropy	56.98	55.10
Amazon	Linear	123	15	60.0	60.0	Entropy	59.03	58.46
Amazon	MLP	123	15	53.3	60.0	Entropy	59.08	58.06
Amazon	Linear	456	15	66.7	66.7	Entropy	58.67	57.36
Amazon	MLP	456	15	66.7	66.7	Entropy	58.12	56.88

A.4 Complete Budget-Allocation Results

The original random-batch policy matches total teacher-query cost but not the cascade's query-depth profile. The additional depth-matched control preserves the cascade's numbers of one-, two-, and three-teacher examples while randomising which examples receive the additional queries.

Table A.8: Cascade versus random-batch allocation at equal total query cost $q = 1.4$. Values are accuracy percentages over five seeds. Δ is cascade minus random; C/R/T gives cascade wins, random wins and ties.

Dataset	L	Cascade	Random batch	Δ (pp)	C/R/T
AG News	10	88.34 ± 0.13	88.62 ± 0.16	-0.28	0/5/0
AG News	50	88.56 ± 0.24	88.75 ± 0.15	-0.19	1/4/0
Yahoo	10	60.86 ± 0.43	60.86 ± 0.34	0.00	2/3/0
Yahoo	50	60.46 ± 0.56	60.32 ± 0.46	$+0.13$	3/2/0
Amazon	10	57.69 ± 1.01	57.43 ± 1.56	$+0.26$	4/1/0
Amazon	50	54.97 ± 1.13	55.11 ± 1.81	-0.14	2/3/0
Amazon-3	10	74.03 ± 0.87	74.23 ± 0.78	-0.20	2/3/0
Amazon-3	50	71.58 ± 1.34	71.59 ± 2.12	-0.01	3/2/0

Table A.9: Cascade versus random-batch allocation at equal total query cost $q = 2.0$. Values are accuracy percentages over seeds 42, 123 and 456. Δ is cascade minus random; C/R/T gives cascade wins, random wins and ties.

Dataset	L	Cascade	Random batch	Δ (pp)	C/R/T
AG News	10	88.38 ± 0.30	88.65 ± 0.13	-0.28	0/3/0
AG News	50	88.43 ± 0.32	88.46 ± 0.20	-0.03	1/2/0
Yahoo	10	61.36 ± 0.61	60.88 ± 0.48	$+0.48$	3/0/0
Yahoo	50	60.15 ± 0.39	60.42 ± 0.26	-0.26	1/2/0
Amazon	10	58.20 ± 0.38	58.48 ± 0.42	-0.28	0/3/0
Amazon	50	53.77 ± 2.13	55.06 ± 2.04	-1.29	0/3/0
Amazon-3	10	73.46 ± 0.37	73.54 ± 0.65	-0.08	1/1/1
Amazon-3	50	71.71 ± 3.32	72.61 ± 1.78	-0.90	1/2/0

Table A.10: Cascade versus depth-matched random allocation. Accuracy values are mean $\pm$ sample standard deviation over seeds $42, 123, 456$. Δ is cascade minus random in percentage points. C/R/T gives the seed-level direction: cascade, random, or tie. $p_{\max}$ is the largest Holm-adjusted McNemar p-value; no row satisfies the robustness criterion.

Dataset	L	q	Cascade	Depth-matched random	Δ	C/R/T	$p_{\max}$
AG News	10	1.4	88.36 ± 0.15	88.54 ± 0.13	-0.19	R/R/R	0.823
AG News	50	1.4	88.46 ± 0.14	88.67 ± 0.12	-0.21	C/R/R	0.673
Yahoo	10	1.4	60.97 ± 0.54	60.95 ± 0.22	$+0.02$	C/R/R	1.000
Yahoo	50	1.4	60.49 ± 0.66	60.71 ± 0.29	-0.23	C/R/R	0.471
Amazon	10	1.4	58.27 ± 0.48	58.29 ± 0.32	-0.02	R/C/R	1.000
Amazon	50	1.4	54.78 ± 1.21	54.07 ± 1.95	$+0.70$	C/C/R	0.664
Amazon-3	10	1.4	73.57 ± 0.83	73.99 ± 0.46	-0.42	C/R/R	0.865
Amazon-3	50	1.4	72.19 ± 1.45	72.99 ± 1.54	-0.80	R/R/C	1.000
AG News	10	2.0	88.38 ± 0.30	88.26 ± 0.23	$+0.12$	C/R/C	1.000
AG News	50	2.0	88.43 ± 0.32	88.47 ± 0.12	-0.04	R/R/C	0.329
Yahoo	10	2.0	61.36 ± 0.61	61.29 ± 0.23	$+0.07$	C/R/R	0.699
Yahoo	50	2.0	60.15 ± 0.39	60.38 ± 0.38	-0.23	C/R/R	0.745
Amazon	10	2.0	58.20 ± 0.38	58.48 ± 0.20	-0.28	R/R/R	1.000
Amazon	50	2.0	53.77 ± 2.13	53.92 ± 1.63	-0.16	R/C/C	0.074
Amazon-3	10	2.0	73.46 ± 0.37	73.32 ± 0.49	$+0.14$	C/C/R	0.534
Amazon-3	50	2.0	71.71 ± 3.32	72.42 ± 2.14	-0.72	C/R/R	0.290

A.5 Sensitivity Analyses

A.5.1 Sensitivity to the Supplied Class Prior

The main experiments use the uniform class prior known from the balanced working pools. To examine dependence on this assumption, the cached teacher logits were re-aggregated under correct, mildly misspecified and strongly misspecified supplied priors. For a misspecification strength s, the assumed prior was

$$\pi^{(s,j)} = (1-s)\pi_{\text{uniform}} + s\,e_j,$$

where e_j places all mass on class j. Results are averaged over every choice of j, with $s = 0.15$ and $s = 0.50$ representing mild and strong misspecification, respectively.

Table A.11: Teacher-target sensitivity to the supplied class prior. Cells report accuracy/NLL/ECE using cached logits. Misspecified-prior results average over each possible dominant class.

Method	AG News	Yahoo	Amazon	Amazon-3
Probability mean	0.879/0.479/0.073	0.658/1.456/0.168	0.634/1.035/0.159	0.759/0.593/0.108
Scale alignment only	0.879/0.427/0.065	0.658/1.269/0.110	0.632/0.909/0.084	0.763/0.536/0.036
Correct uniform prior	**0.899**/**0.343**/**0.022**	**0.663**/**1.184**/0.088	**0.659**/**0.851**/**0.026**	**0.801**/**0.473**/**0.026**
Mildly misspecified prior	0.869/0.533/0.068	0.642/1.339/**0.081**	0.635/0.990/0.059	0.784/0.553/0.035
Strongly misspecified prior	0.650/1.032/0.137	0.458/1.941/0.212	0.439/1.413/0.206	0.618/0.945/0.154

The correct prior gives the highest target accuracy and lowest NLL on every task. Mild misspecification already reduces accuracy relative to the correct prior, while strong misspecification causes substantial deterioration. The benefit of prior matching is therefore conditional on reliable prior information. This target-level analysis averages over all possible dominant classes; Section 4.3.1 reports the corresponding student experiment for one predeclared direction. Both analyses vary the supplied prior on balanced pools and do not test covariate shift or a new deployment domain.

A.5.2 Subset-Only Cascade Alignment

The principal cascade estimates every teacher's alignment parameters from the complete cached pool. A deployment-compatible alternative would estimate the parameters of T_2 and T_3 only from the subsets on which they were queried. Table A.12 compares these choices using cached logits while holding the escalation sets fixed.

Table A.12: Teacher-target sensitivity to subset-only cascade alignment at $q = 1.4$. Accuracy differences are subset-only minus full-pool alignment. The analysis uses cached logits and does not retrain students.

Dataset	Full-pool accuracy/NLL	Subset-only accuracy/NLL	Δ accuracy (pp)
AG News	0.8992/0.4079	0.8920/0.4325	−0.72
Yahoo	0.6612/1.3301	0.6608/1.3914	−0.04
Amazon	0.6596/1.1117	0.6548/1.1029	−0.48
Amazon-3	0.7993/0.5870	0.7907/0.6152	−0.87

Subset-only estimation lowers teacher-target accuracy by 0.04–0.87 percentage points across the four configurations. The full-pool estimates are therefore mildly optimistic. Because the escalation sets are held fixed and no students are retrained, this analysis does not constitute a complete online-cascade evaluation.

A.6 Relationship to Existing Methods

Table A.13 distinguishes methods reproduced in the experiments from output-level adaptations and methods discussed only as related work. Published accuracy values are not compared directly because the original studies use different datasets, teacher access, student architectures and supervision assumptions.

Table A.13: Implementation status and relationship of published distillation methods to the evaluated pipeline.

Method	Required information	Gold-label use	Status	Relationship to this thesis
You et al. [35]	Teacher outputs and intermediate representations	Labelled student training	Output component only	Probability averaging reproduces the output-level component; the intermediate-representation constraints are unavailable for black-box teachers.
Kwon et al. [18]	Teacher soft labels and their entropy	None for teacher selection	Inspired adaptation	The paper uses entropy to select a teacher, whereas we applied continuous exponential weights and aggregates teacher logits.
CA-MKD [38]	Teacher predictions and per-example labels	Required	Not implemented	Determines sample-wise teacher weights from their agreement with the ground-truth label, which is unavailable for the unlabelled working pool.
RL-KD [36]	Input representations, teacher probabilities and teacher losses	Required	Not implemented	Selects teachers per instance through joint RL and student training; unlike our router, it uses label-dependent states and delayed student-performance rewards.
DE-MKD [5]	Teacher outputs, intermediate features and decoupled losses	Labelled student training	Not implemented	The entropy heuristic does not reproduce the paper's decoupled target/non-target and feature-transfer objectives.
AE-KD [8]	Teacher-specific student gradients	Labelled student training	Not implemented	Resolves teacher conflict through multi-objective optimisation in gradient space.
GOVERN [44]	Teacher outputs and student-gradient orientations	None during unsupervised KD	Not implemented	Performs dynamic teacher selection through gradient-orientation voting rather than teacher-prior alignment.
PILE [3]	Teacher logits and labelled guidance	Required	Not implemented	Constructs a supervised iterative logit ensemble and therefore uses more label information than the present setting permits.
TinyLLM [32]	Generated answers, rationales and in-context examples	Correct-answer supervision	Related work	Distils reasoning into a generative LLM student rather than class probabilities into an encoder classifier.
M-RARU [20]	Student uncertainty and sequential teacher queries	None for query selection	Related work	Allocates single-teacher queries across examples, whereas this thesis allocates additional teachers to selected examples.
Prior-and-scale alignment	Cached teacher class logits and supplied class prior	No individual labels; prior supplied	Implemented	Corrects class-marginal and confidence-scale differences before probability aggregation.
Entropy cascade	Sequential aligned teacher probabilities	None for allocation	Implemented	Allocates a fixed multi-teacher query budget using first-teacher entropy and inter-teacher disagreement.

A.7 Supervised-Baseline Sensitivity

To test whether the fixed configuration understates direct supervision, learning rate and training duration were selected using only $\mathcal{D}_L$. Stratified five-fold cross-validation considered $\eta \in \{1,2,5\} \times 10^{-5}$ and $E \in \{2,3,5\}$, selecting the configuration with the highest mean macro-F1 across folds and labelled-set seeds. The selected configuration was then retrained on all labelled examples for each seed; test labels were not used for selection.

All eight cells selected $E = 5$. Seven selected $\eta = 5 \times 10^{-5}$; Amazon at $L = 10$ selected $\eta = 2 \times 10^{-5}$.

Table A.14: Supervised-baseline sensitivity. Accuracies are percentages reported as mean ± sample standard deviation over seeds $42, 123, 456$. Δ is raw-Qwen KD minus the CV-selected supervised baseline.

Dataset	L	Fixed supervised	CV-selected supervised	Raw-Qwen KD	Δ
AG News	10	65.09 ± 5.77	72.00 ± 1.90	87.14 ± 0.49	$+15.13$
AG News	50	84.42 ± 1.30	85.16 ± 1.60	87.08 ± 0.10	$+1.93$
Yahoo	10	26.76 ± 2.66	40.83 ± 4.50	59.20 ± 0.57	$+18.38$
Yahoo	50	52.14 ± 1.41	54.50 ± 0.99	58.70 ± 1.45	$+4.20$
Amazon	10	22.00 ± 1.18	26.05 ± 1.63	57.35 ± 0.82	$+31.30$
Amazon	50	33.76 ± 2.35	36.41 ± 2.53	57.32 ± 1.05	$+20.92$
Amazon-3	10	38.88 ± 1.67	40.57 ± 0.46	72.11 ± 0.99	$+31.53$
Amazon-3	50	47.66 ± 2.81	57.64 ± 3.01	73.76 ± 0.24	$+16.12$

Raw-Qwen KD has the higher mean in every cell and a consistent positive direction across all seed-level comparisons. Seven cells satisfy the complete robustness criterion. On AG News at $L = 50$, the advantage is positive for all three seeds, but the seed-456 comparison is not significant after Holm adjustment ($p_{\text{Holm}} = 0.0849$).

A.8 Confidence-Scale Diagnostics

Table A.15 reports the temperatures fitted after prior matching and the resulting target entropies. To isolate the incremental effect of confidence-scale alignment, H_{prior} is measured after prior matching but before scale alignment. Mistral is the least-confident reference teacher and therefore has $t_{\text{Mistral}} = 1$ throughout.

Table A.15: Confidence-scale diagnostics. Entropies are in nats. H_{scale} is measured after prior matching and scale alignment; H_{KD} additionally applies the KD temperature $\tau = 2$.

Dataset	t_{Qwen}	t_{Gemma}	H_{prior}	H_{scale}	H_{KD}
AG News	1.391	2.561	0.164	0.245	0.689
Yahoo	1.241	3.160	0.604	0.838	1.726
Amazon	2.024	4.802	0.563	0.868	1.309
Amazon-3	1.980	4.506	0.334	0.517	0.854

Scale alignment increases target entropy on every task. The largest absolute increase occurs on Amazon, from 0.563 to 0.868 nats, where the student-level transfer reversal is also strongest. This supports, but does not establish, the proposed over-softening mechanism.

Appendix B

Algorithms

B.1 Teacher Logit Extraction

Algorithm 1 Guided Teacher-Logit Extraction

Require: Working pool $\mathcal{D} = \{x_i\}_{i=1}^{N}$; fixed few-shot demonstrations $\mathcal{E}$; teacher T_k; single-token class verbalizers $\mathcal{V} = \{v_1, \ldots, v_C\}$; KD temperature $\tau > 0$

Ensure: Cached class-logit matrix $Z_k \in \mathbb{R}^{N \times C}$ and soft-label matrix $P_k^{(\tau)} \in \mathbb{R}^{N \times C}$

for $i = 1$ **to** N **do**
  $prompt_i \leftarrow \textsc{BuildPrompt}(x_i, \mathcal{E}, \mathcal{V})$
  $z_k(x_i) \leftarrow \textsc{GuidedLogProbs}(T_k, prompt_i, \mathcal{V})$ ▷ one score for every class token
  $p_k^{(\tau)}(x_i) \leftarrow \mathrm{softmax}(z_k(x_i)/\tau)$
end for
$Z_k \leftarrow [z_k(x_1), \ldots, z_k(x_N)]^\top$
$P_k^{(\tau)} \leftarrow [p_k^{(\tau)}(x_1), \ldots, p_k^{(\tau)}(x_N)]^\top$
$\textsc{Cache}(Z_k, P_k^{(\tau)})$ ▷ indexed by immutable sample identifier
return $Z_k, P_k^{(\tau)}$

Labels in $\mathcal{D}$ are not used during extraction. The few-shot demonstrations in $\mathcal{E}$ are counted inside the declared label budget. Although the returned values are log-probabilities over the constrained verbalizer set, they can be treated as class logits because softmax is invariant to a shared additive constant.

B.2 Prior-and-Scale Alignment under a Supplied Class Prior

Algorithm 2 uses no individual labels while fitting b_k or t_k, but it assumes access to the target class prior π. In the main experiments, π is known to be uniform because the working pools were constructed to be class-balanced. We therefore describe the method as unsupervised conditional on a supplied prior, rather than label-free calibration.

B.3 Router Fitting with LOOCV Model Selection and Safety Gate

Table B.1: Disagreement features used by the learned router.

#	Feature	Definition
1	$H(\bar{p}(x))$	consensus entropy
2	$\overline{D}(x)$	mean pairwise symmetrised KL divergence
3	$D_{\max}(x)$	maximum pairwise symmetrised KL divergence
4	$A_1(x)$	fraction of teacher pairs with the same top-1 class
5	$\bar{\kappa}(x)$	mean pairwise Kendall's τ between class rankings
6–8	$v_m(x)$	variance of $p_{k,c_m}(x)$ across teachers, $m = 1,2,3$
9–11	$\delta_k(x)$	divergence of teacher k from the consensus

Algorithm 2 Prior-and-Scale Alignment under a Supplied Class Prior

Require: Cached teacher logits $\{Z_k\}_{k=1}^{K}$ over $\mathcal{D}$; supplied class prior $\pi \in \Delta^{C-1}$ with $\pi_c > 0$; KD temperature τ; maximum iterations M; tolerance ε

Ensure: Purified KD targets $P_T \in \mathbb{R}^{N\times C}$; alignment parameters $\{b_k, t_k\}_{k=1}^{K}$

Stage 1: class-prior matching

for $k = 1$ **to** K **do**
  $b_k \leftarrow \mathbf{0} \in \mathbb{R}^C$
  for $m = 1$ **to** M **do**
    $\hat{\pi}_k \leftarrow \dfrac{1}{N}\sum_{x\in\mathcal{D}} \mathrm{softmax}(z_k(x) + b_k)$
    $\Delta b_k \leftarrow \log \pi - \log\big(\max(\hat{\pi}_k, 10^{-12})\big)$ ▷ element-wise operations
    $b_k \leftarrow b_k + \Delta b_k$
    if $\|\Delta b_k\|_\infty < \varepsilon$ **then**
      break
    end if
  end for
end for

Stage 2: confidence-scale alignment

for $k = 1$ **to** K **do**
  $H_k \leftarrow \dfrac{1}{N}\sum_{x\in\mathcal{D}} H(\mathrm{softmax}(z_k(x) + b_k))$
end for
$H^\star \leftarrow \max_k H_k$ ▷ entropy of the least-confident teacher
for $k = 1$ **to** K **do**
  find $t_k \in [0.25, 16]$ by bisection such that

$$\left| \frac{1}{N}\sum_{x\in\mathcal{D}} H\left(\mathrm{softmax}\left(\frac{z_k(x) + b_k}{t_k}\right)\right) - H^\star \right| < \varepsilon$$

end for

Stage 3: probability-space aggregation

for each $x \in \mathcal{D}$ **do**
  for $k = 1$ **to** K **do**
    $\tilde{p}_k(x) \leftarrow \mathrm{softmax}\left(\dfrac{z_k(x) + b_k}{t_k}\right)$
  end for
  $\bar{p}(x) \leftarrow \dfrac{1}{K}\sum_{k=1}^{K} \tilde{p}_k(x)$
  $P_T[x] \leftarrow \mathrm{softmax}\left(\dfrac{\log \max(\bar{p}(x), 10^{-12})}{\tau}\right)$ ▷ KD softening
end for
return $P_T, \{b_k, t_k\}_{k=1}^{K}$

Algorithm 3 Router Fitting with LOOCV Model Selection and Safety Gate

Require: Teacher logits $\{z_k(x)\}_{k=1}^{K}$; features $\phi(x)$ for all $x \in \mathcal{D}$; calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^{n_{\text{cal}}}$; router architecture $r \in \{\text{LINEAR}, \text{MLP}\}$; weight-decay grid Λ; temperature τ

Ensure: Teacher targets P_T; gate decision $g \in \{\text{ROUTER}, \text{MEAN}, \text{ENTROPY}\}$

$(\mu_\phi, \sigma_\phi) \leftarrow$ population mean and standard deviation of $\{\phi(x) : x \in \mathcal{D}\}$
for each $x \in \mathcal{D}$ **do**
  $\tilde{\phi}(x) \leftarrow \dfrac{\phi(x) - \mu_\phi}{\max(\sigma_\phi, 10^{-8})}$
  for $k = 1$ **to** K **do**
    $p_k(x) \leftarrow \text{softmax}(z_k(x))$
  end for
  $w_k^{\text{prior}}(x) \leftarrow \dfrac{\exp[-H(p_k(x))]}{\sum_{j=1}^{K} \exp[-H(p_j(x))]} \quad \forall k$
end for
for each $\lambda \in \Lambda$ **do**
  $a_\lambda \leftarrow \text{LOOCVACCURACY}(r, \lambda, \mathcal{D}_{\text{cal}}, \tilde{\phi}, z_{1:K}, w^{\text{prior}}, \tau)$ ▷ router retrained in every fold
end for
$\lambda^\star \leftarrow \arg\max_{\lambda \in \Lambda} a_\lambda$
$a_{\text{router}} \leftarrow a_{\lambda^\star}$
$a_{\text{mean}} \leftarrow$ calibration accuracy of the raw-logit mean on $\mathcal{D}_{\text{cal}}$
$a_{\text{ent}} \leftarrow$ calibration accuracy of the entropy-weighted logit mean on $\mathcal{D}_{\text{cal}}$
if $a_{\text{router}} > \max(a_{\text{mean}}, a_{\text{ent}})$ **then**
  $\theta_r^\star \leftarrow \text{FITROUTER}(r, \lambda^\star, \mathcal{D}_{\text{cal}}, \tilde{\phi}, z_{1:K}, w^{\text{prior}}, \tau)$
  for each $x \in \mathcal{D}$ **do**
    $\delta(x) \leftarrow r_{\theta_r^\star}(\tilde{\phi}(x))$
    $w(x) \leftarrow \text{softmax}\big(\delta(x) + \log w^{\text{prior}}(x)\big)$
    $P_T[x] \leftarrow \text{softmax}\left(\dfrac{\sum_{k=1}^{K} w_k(x) z_k(x)}{\tau}\right)$
  end for
  $g \leftarrow$ ROUTER
else if $a_{\text{ent}} \geq a_{\text{mean}}$ **then**
  $P_T \leftarrow P_{\text{ent}}$
  $g \leftarrow$ ENTROPY
else
  $P_T \leftarrow P_{\text{mean}}$
  $g \leftarrow$ MEAN
end if
return P_T, g

B.4 Offline Evaluation of an Entropy-Based Query Cascade

Algorithm 4 Offline Evaluation of an Entropy-Based Query Cascade

Require: Cached logits $\{Z_k\}_{k=1}^{3}$ over $\mathcal{D}$; teacher order (T_1, T_2, T_3); supplied class prior π; requested deployment query rate $q \in [1,3]$; requests above the attainable ceiling $2 + d_{12}$ saturate; KD temperature τ

Ensure: Cascade targets P_T; realised query rate $\hat{q}$

obtain aligned teacher distributions $\tilde{p}_1, \tilde{p}_2, \tilde{p}_3$ and parameters $\{b_k, t_k\}_{k=1}^{3}$ using Algorithm 2

for each $x \in \mathcal{D}$ **do**

    $e_1(x) \leftarrow H(\tilde{p}_1(x))$

end for

choose entropy threshold h by bisection to minimise

$$\left| 1 + \frac{|\mathcal{S}_2(h)|}{N} + \frac{|\mathcal{S}_3(h)|}{N} - q \right|,$$

where $\mathcal{S}_2(h) = \{x : e_1(x) > h\}$ and

$$\mathcal{S}_3(h) = \{x \in \mathcal{S}_2(h) : \arg\max \tilde{p}_1(x) \neq \arg\max \tilde{p}_2(x)\}$$

$$\hat{q} \leftarrow 1 + \frac{|\mathcal{S}_2(h)|}{N} + \frac{|\mathcal{S}_3(h)|}{N}$$

for each $x \in \mathcal{D}$ **do**

    if $x \notin \mathcal{S}_2(h)$ **then**

        $\bar{p}(x) \leftarrow \tilde{p}_1(x)$ ▷ one hypothetical query

    else if $x \notin \mathcal{S}_3(h)$ **then**

        $\bar{p}(x) \leftarrow \frac{1}{2}(\tilde{p}_1(x) + \tilde{p}_2(x))$ ▷ two hypothetical queries

    else

        $\bar{p}(x) \leftarrow \frac{1}{3}\sum_{k=1}^{3} \tilde{p}_k(x)$ ▷ three hypothetical queries

    end if

    $P_T[x] \leftarrow \mathrm{softmax}\left(\frac{\log \max(\bar{p}(x), 10^{-12})}{\tau} \right)$

end for

return $P_T, \hat{q}$

All teacher logits were precomputed to permit reproducible offline comparison of allocation policies. Consequently, $\hat{q}$ is the number of calls the cascade would make under prospective deployment, rather than the number of calls made while constructing the experimental cache. The principal cascade uses alignment parameters estimated

from the full cached pools. A cached-logit subset-alignment analysis in Table A.12 quantifies the teacher-target effect of this choice while holding the escalation sets fixed.

B.5 Student Training with Soft and Hard Supervision

Algorithm 5 Student Training with Soft and Hard Supervision

Require: Student $f_S(\theta)$; training pool $\mathcal{D}_{\text{train}}$; KD targets P_T over $\mathcal{D}_{\text{train}}$ when $\alpha > 0$; CE-labelled subset $\mathcal{D}_{\text{CE}}$; KD weight $\alpha \in [0,1]$; temperature τ; learning rate η; epochs E

Ensure: Trained parameters $\theta^\star$

for $e = 1$ **to** E **do**
  for each mini-batch $\mathcal{B} \subseteq \mathcal{D}_{\text{train}}$ **do**
    $z_S(x) \leftarrow f_S(x;\theta) \quad \forall x \in \mathcal{B}$
    $p_S^{(\tau)}(x) \leftarrow \text{softmax}(z_S(x)/\tau) \quad \forall x \in \mathcal{B}$
    if $\alpha > 0$ **then**
      $\mathcal{L}_{\text{KD}} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} D_{\text{KL}}\left(P_T[x] \,\|\, p_S^{(\tau)}(x)\right)$
    else
      $\mathcal{L}_{\text{KD}} \leftarrow 0$
    end if
    $\mathcal{B}_{\text{CE}} \leftarrow \mathcal{B} \cap \mathcal{D}_{\text{CE}}$
    if $|\mathcal{B}_{\text{CE}}| > 0$ **then**
      $\mathcal{L}_{\text{CE}} \leftarrow \frac{1}{|\mathcal{B}_{\text{CE}}|} \sum_{(x,y) \in \mathcal{B}_{\text{CE}}} \mathcal{L}_{\text{CE}}(y, z_S(x))$
    else
      $\mathcal{L}_{\text{CE}} \leftarrow 0$
    end if
    $\mathcal{L} \leftarrow \alpha\tau^2 \mathcal{L}_{\text{KD}} + (1-\alpha)\mathcal{L}_{\text{CE}}$
    $\theta \leftarrow \textsc{AdamWUpdate}(\theta, \nabla_\theta \mathcal{L}, \eta)$
  end for
end for
$\theta^\star \leftarrow \theta$
return $\theta^\star$

For the supervised baseline, $\mathcal{D}_{\text{train}} = \mathcal{D}_{\text{CE}} = \mathcal{D}_L$ and $\alpha = 0$. For fixed-rule KD baselines, $\mathcal{D}_{\text{train}} = \mathcal{D}$ and $\mathcal{D}_{\text{CE}} = \mathcal{D}_L$. For learned-router experiments, $\mathcal{D}_{\text{train}} = \mathcal{D}$ and $\mathcal{D}_{\text{CE}} = \mathcal{D}_{L\text{-tr}}$

because $\mathcal{D}_{\text{cal}}$ is reserved for router training and model selection. Calibration examples remain in the KD pool and receive soft-target supervision, but they do not contribute a hard-label cross-entropy gradient.

Appendix C

Teacher Prompt Template

The following schematic shows the prompt structure used for teacher-logit extraction. Category names and the number of demonstrations varied by dataset.

System:

You are a strict single-token text classifier.

The only valid outputs are: A, B, ..., [final verbalizer].

Output exactly one uppercase character, without explanation.

Categories:

A: [class name 1]

B: [class name 2]

...

[final verbalizer]: [final class name]

User:

Text: ``[demonstration from class 1]''

Assistant:

A

User:

Text: ``[demonstration from class 2]''

Assistant:

B

...

User:

Text: ``[input text]''

Assistant:

Appendix D

ModernBERT Configurations

Table D.1: Student training configuration. All values are shared by every method and every seed; $\alpha = 0$ recovers the supervised baseline and $\mathcal{D}_{\text{CE}} = \mathcal{D}_{L\text{-tr}}$ replaces $\mathcal{D}_L$ in router runs.

Setting	Value
Student	ModernBERT-base [33]
Parameters	$\approx$ 149.6M (including task-specific head)
Maximum sequence length	256 tokens
Classification head	Attention-mask-aware mean pooling, followed by $768 \to 768$ dense layer, GELU, LayerNorm, and $768 \to C$ linear classifier
Optimiser	AdamW
Learning rate η	2×10^{-5}
Schedule	Constant learning rate; no warmup
Weight decay	0.01
Batch size b	16
Epochs E	3
Precision	FP32
KD weight α	0.5 (0 for supervised)
Temperature τ	2.0
KD applied to	$\mathcal{D}_L \cup \mathcal{D}_U$
CE applied to	$\mathcal{D}_L$ ($\mathcal{D}_{L\text{-tr}}$ in router runs)
Seeds	$42, 123, 456$ (plus $789, 2024$ for matched-budget runs)
Student-training hardware	NVIDIA L40S GPU (46 GB)
Approximate runtime per run	0.5 min (supervised); 1.7 min (KD)